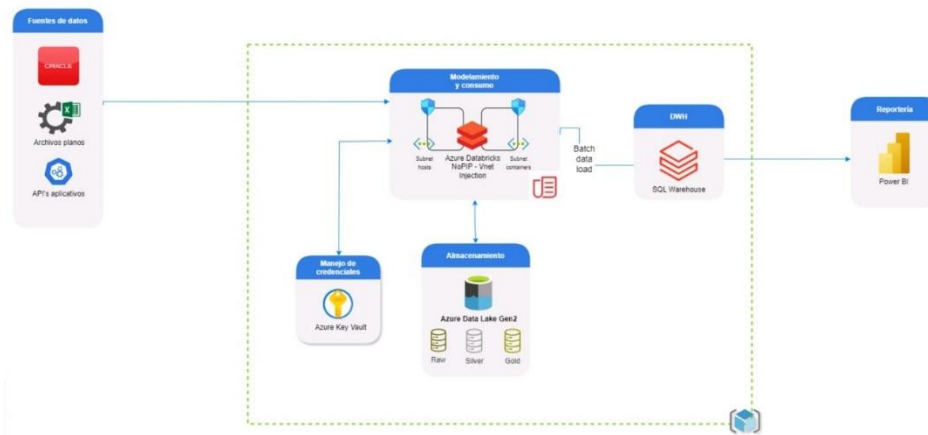


Despliegue Lakehouse con Databricks en Azure



Esta guía es el paso a paso para desplegar una arquitectura Lakehouse usando Databricks desde Azure.

Adicionalmente, contiene un ejercicio práctico que consiste en la realización de un pipeline de extracción y transformación de datos, orquestado. Lo anterior, teniendo en cuenta el uso de la arquitectura medallón.

Contenido

Parte 1:

- Creación de grupo de recursos.
- Creación de la cuenta de almacenamiento.
- Creación de access connector.
- Creación de Key vault.
- Creación de espacio de trabajo de Databricks.
- Permisos de conexión a la cuenta de almacenamiento desde el Access connector de Databricks.
- Permisos de usuario sobre la suscripción(opcional).

Parte 2:

- Despliegue Cluster de Databricks.
- Crear credencial de almacenamiento.
- Creación de external location.
- Creación de Catálogos.
- Otorgar permisos a los external location.
- Otorgar permisos a los catálogos.
- Definición de grupos de trabajo (opcional).
- Creación del Scope.
- Creación de carpeta de trabajo.

Parte 3

- Importación de notebooks de extracción y transformación y de orquestación
- Creación de directorios para el almacenamiento de datos
- Creación de esquema de clientes.
- Programación de Job.

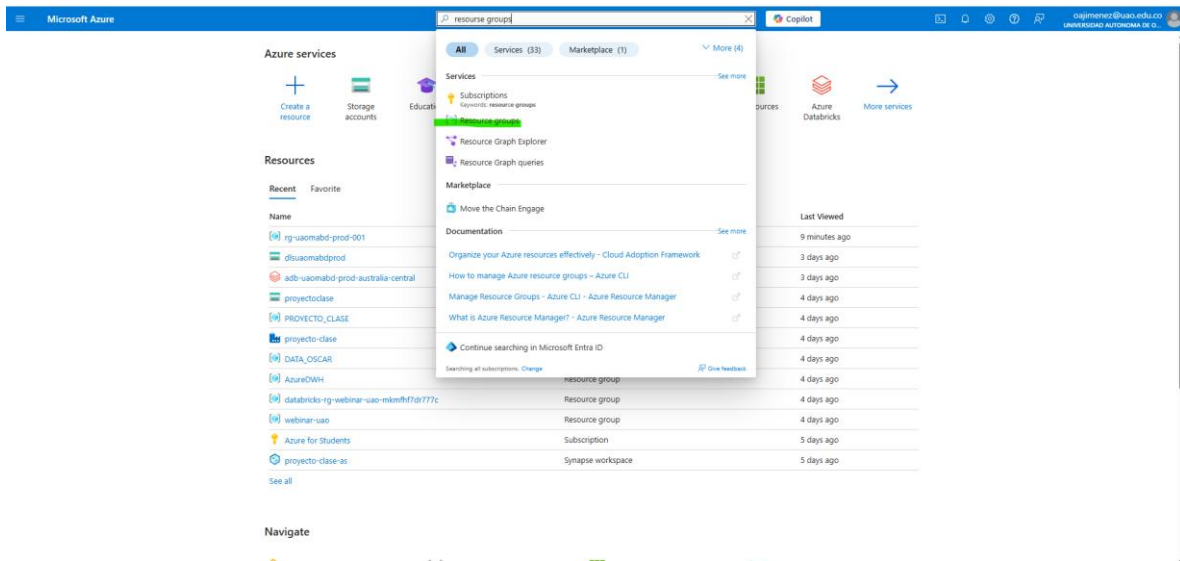
Nota: Para nombrar los recursos haremos uso de la siguiente guía de Azure:

<https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/azure-best-practices/resource-naming>

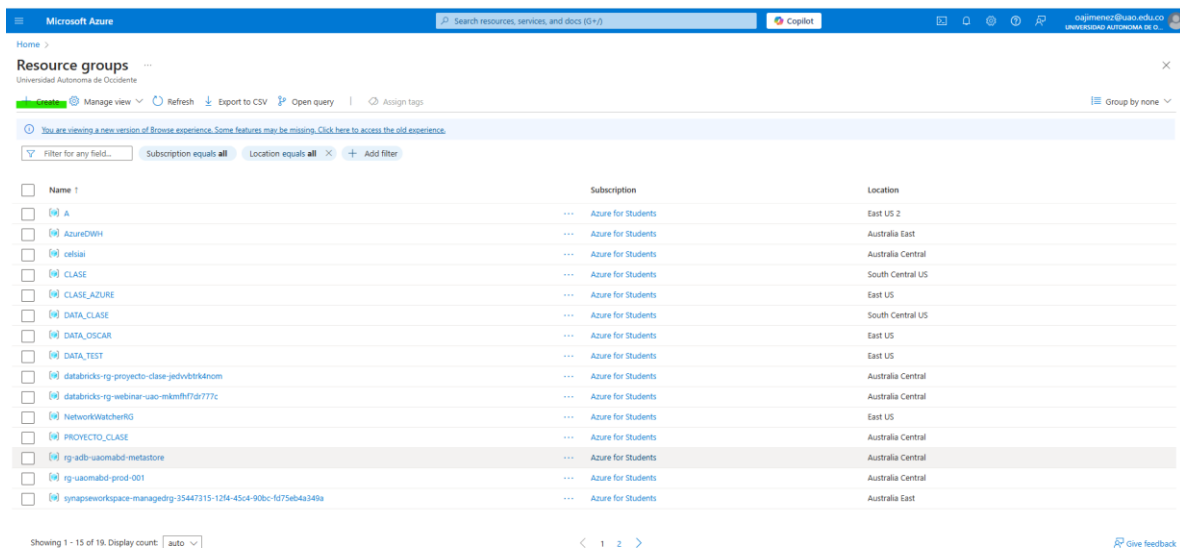
Parte 1

Creación de grupo de recursos

1. Desde la ventana de home, buscar por “Grupo de Recursos”



2. Hacemos clic en **crear**



3. Creamos el grupo de recursos con la siguiente información:

- Nombre del grupo de recursos: rg-uao-prod-001
- Región: Australia Central

Home > Resource groups >

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription ⓘ Azure for Students

Resource group name ⓘ rg-uao-prod-001

Region ⓘ (Asia Pacific) Australia Central

Previous Next Review + create

4. Dar clic en siguiente hasta dar llegar a **crear**.

Microsoft Azure Search resources, services, and docs (G+)

Home > Resource groups >

Create a resource group

Basics Tags Review + create

[Automation Link](#)

Basics

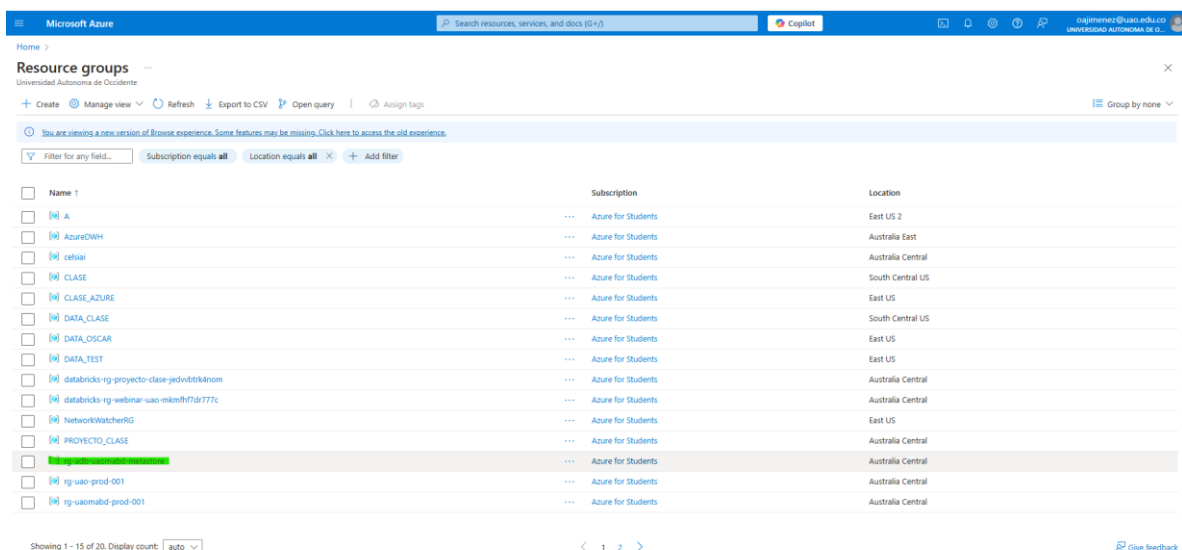
Subscription	Azure for Students
Resource group name	rg-uao-prod-001
Region	Australia Central

Tags

None

Previous Next Create

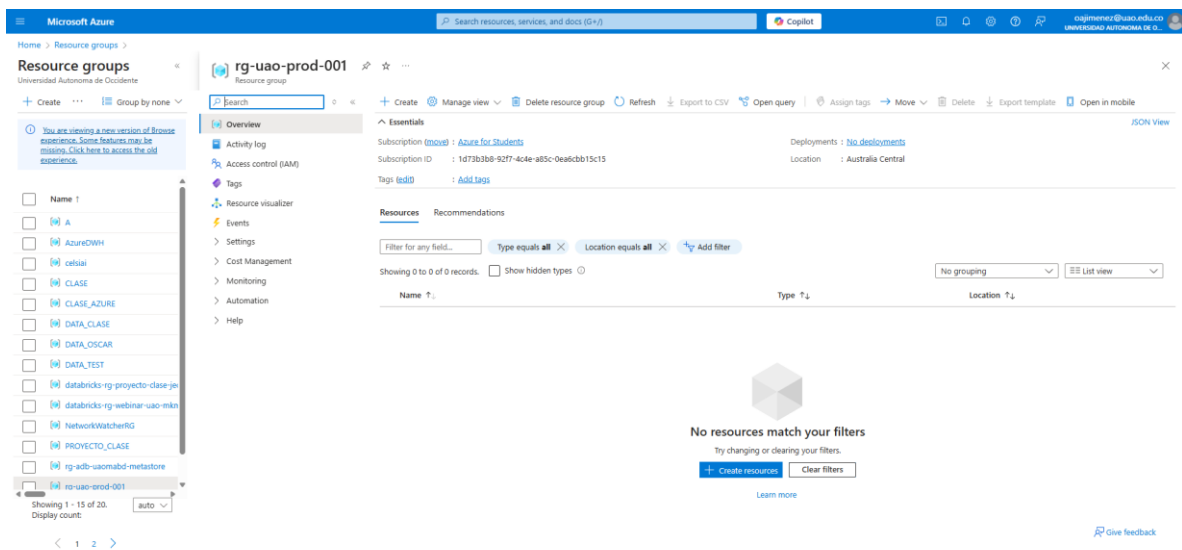
5. Una vez creado actualizamos la página y debería estar disponible.



The screenshot shows the Microsoft Azure portal interface. At the top, there's a search bar and a 'Copilot' button. Below the header, the 'Resource groups' section is active. A message indicates a new version of the Browse experience is available. A filter bar shows 'Subscription equals all' and 'Location equals all'. A table lists 20 resource groups, with 'rg-uao-prod-001' highlighted in green. The table columns are Name, Subscription, and Location. The bottom of the page shows 'Showing 1 - 15 of 20. Display count: auto' and a 'Give feedback' link.

Name	Subscription	Location
A	Azure for Students	East US 2
AzureDWH	Azure for Students	Australia East
celai	Azure for Students	Australia Central
CLASE	Azure for Students	South Central US
CLASE_AZURE	Azure for Students	East US
DATA_CLASE	Azure for Students	South Central US
DATA_OSCAR	Azure for Students	East US
DATA_TEST	Azure for Students	East US
databricks-rg-proyecto-clase-jedvtrikn	Azure for Students	Australia Central
databricks-rg-webinar-uao-mkmthf7d777c	Azure for Students	Australia Central
NetworkWatcherRG	Azure for Students	East US
PROYECTO_CLASE	Azure for Students	Australia Central
rg-uao-prod-001	Azure for Students	Australia Central
rg-uaoabd-prod-001	Azure for Students	Australia Central

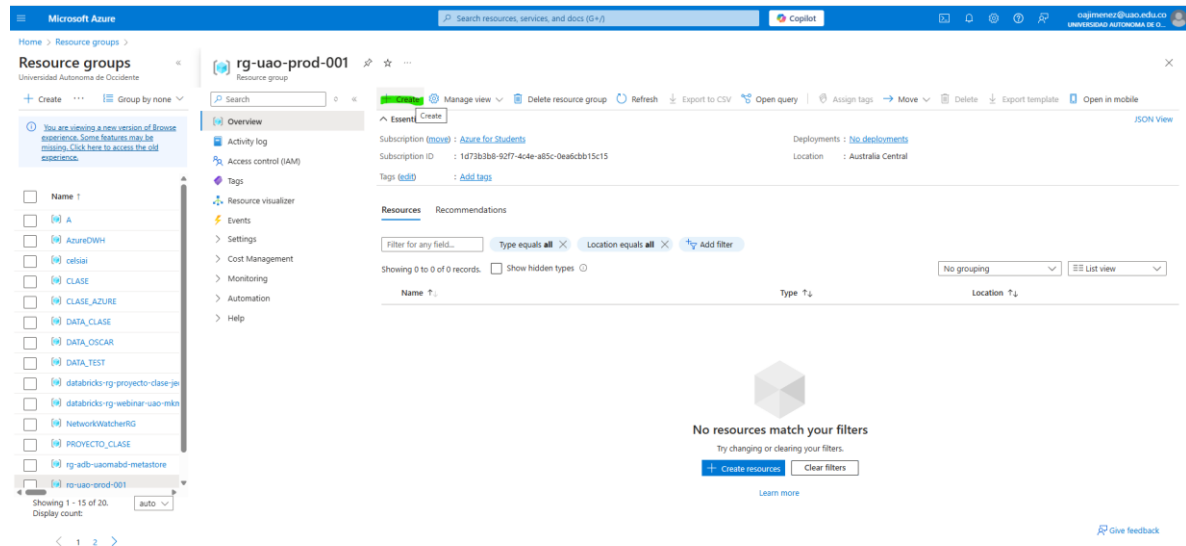
6. Ingresamos a nuestro grupo de recursos.



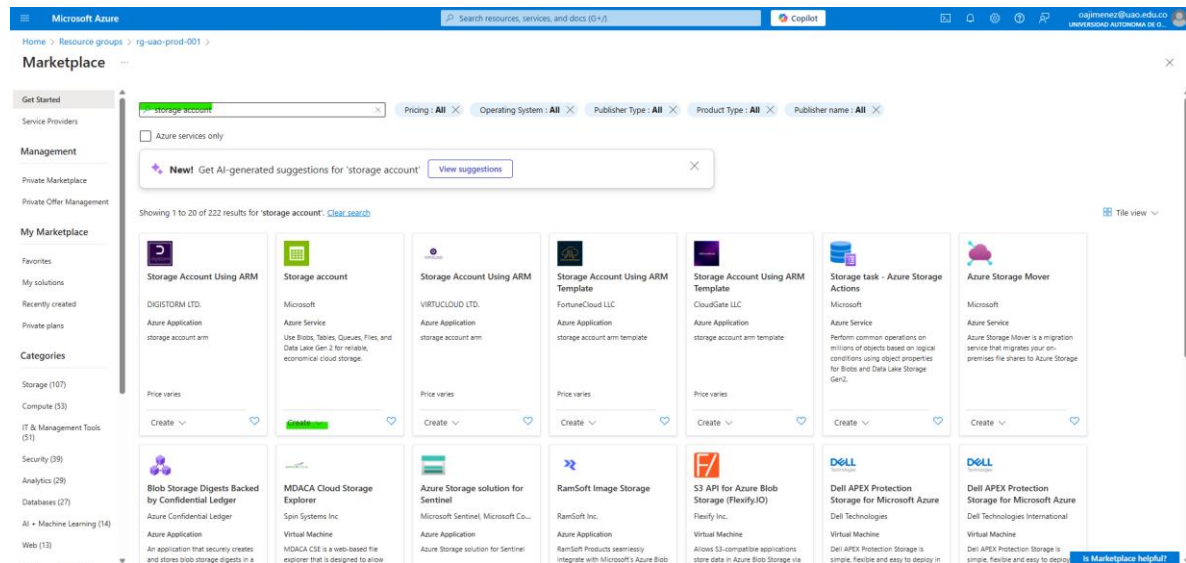
The screenshot shows the details of the 'rg-uao-prod-001' resource group in the Microsoft Azure portal. The left sidebar shows the 'Resource groups' list with 'rg-uao-prod-001' selected. The main area displays the 'Overview' tab for this resource group. It shows the subscription 'Azure for Students', subscription ID '1d73b3b8-927f-4c4e-a85c-0ealcbb15c15', and location 'Australia Central'. There are no deployments or resources listed. A message at the bottom states 'No resources match your filters' with options to 'Create resources' or 'Clear filters'. The bottom of the page shows 'Showing 1 - 15 of 20. Display count: auto' and a 'Give feedback' link.

Creación de la cuenta de almacenamiento

1. Dentro del grupo de recursos damos clic en crear



2. Buscamos cuentas de almacenamiento y damos clic en crear en la opción resaltada



3. Definimos los siguientes parámetros:

Grupo de recursos: El que ya está creado previamente rg-uao-prod-001

- Nombre de la cuenta de almacenamiento: stuaoprod001
- Región: Australia Central
- Primary service: Por defecto
- Performance: Standard
- Redundancia: Locally-redundant-storage(LRS).

Luego de llenar los parametros , damos clic en siguiente.

The screenshot shows the 'Create a storage account' wizard in the Microsoft Azure portal. The 'Basics' tab is selected. The 'Project details' section shows the subscription as 'Azure for Students' and the resource group as 'rg-uao-prod-001'. The 'Instance details' section shows the storage account name as 'stuaoprod001', the region as 'Asia Pacific Australia Central', and the primary service as 'Select a primary service'. The 'Performance' section has 'Standard' selected, and the 'Redundancy' section has 'Locally-redundant storage (LRS)' selected. At the bottom, there are buttons for 'Previous', 'Next', and 'Review + create'.

4. Activamos la casilla de Hierarchical Namespace.

The screenshot shows the 'Create a storage account' wizard in the Microsoft Azure portal, with the 'Advanced' tab selected. The 'Security' section has 'Require secure transfer for REST API operations' checked. The 'Hierarchical Namespace' section has 'Enable hierarchical namespace' checked. The 'Access protocols' section has 'Enable SFTP' and 'Enable network file system v3' unchecked. At the bottom, there are buttons for 'Previous', 'Next', and 'Review + create'.

5. Avanzamos hasta el final y damos clic en crear.

The screenshot shows the 'Create a storage account' page in the Microsoft Azure portal. The 'Review + create' tab is selected. The page displays configuration details for a storage account named 'stuaoprod001' in the 'Australia Central' region. The configuration includes subscription, resource group, location, performance, replication, and security settings. At the bottom, there are 'Previous', 'Next', and 'Create' buttons.

Section	Property	Value
Basics	Subscription	Azure for Students
	Resource group	rg-uao-prod-001
	Location	Australia Central
	Storage account name	stuaoprod001
	Primary service	Standard
Performance	Performance	Standard
	Replication	Locally-redundant storage (LRS)
Advanced	Enable hierarchical namespace	Enabled
	Enable SFTP	Disabled
	Enable network file system v3	Disabled
	Allow cross-tenant replication	Disabled
	Access tier	Hot
	Enable large file shares	Enabled
	Security	Secure transfer
Security	Blob anonymous access	Disabled
	Allow storage account key access	Enabled
	Default to Microsoft Entra authorization in the Azure portal	Disabled

6. Una vez creada nuestra cuenta de almacenamiento ingresamos.

The screenshot shows the 'Overview' page for a storage account named 'stuaoprod001_1742429756221'. The page displays deployment details, including the deployment name, subscription, resource group, start time, and correlation ID. A 'Go to resource' button is visible. The page also includes a sidebar with navigation options (Overview, Inputs, Outputs, Template) and a right-hand panel with recommendations for Cost Management, Microsoft Defender for Cloud, and Free Microsoft tutorials.

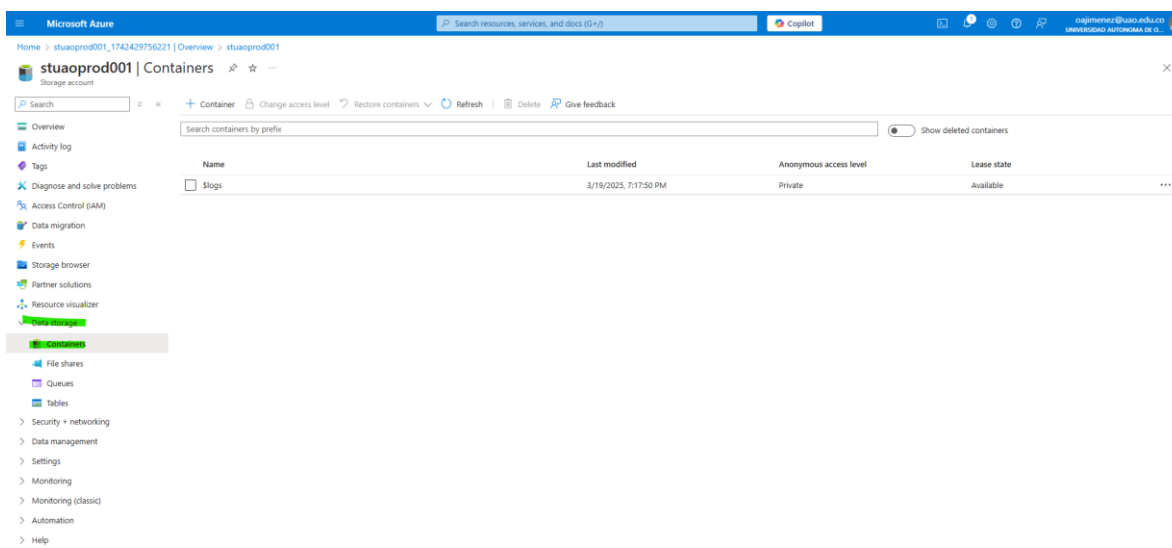
Deployment details:

- Deployment name: stuaoprod001_1742429756221
- Subscription: Azure for Students
- Resource group: rg-uao-prod-001
- Start time: 3/19/2023, 7:16:49 PM
- Correlation ID: ba6646da-ff06-4c7c-8730-4133b7d6b7aa

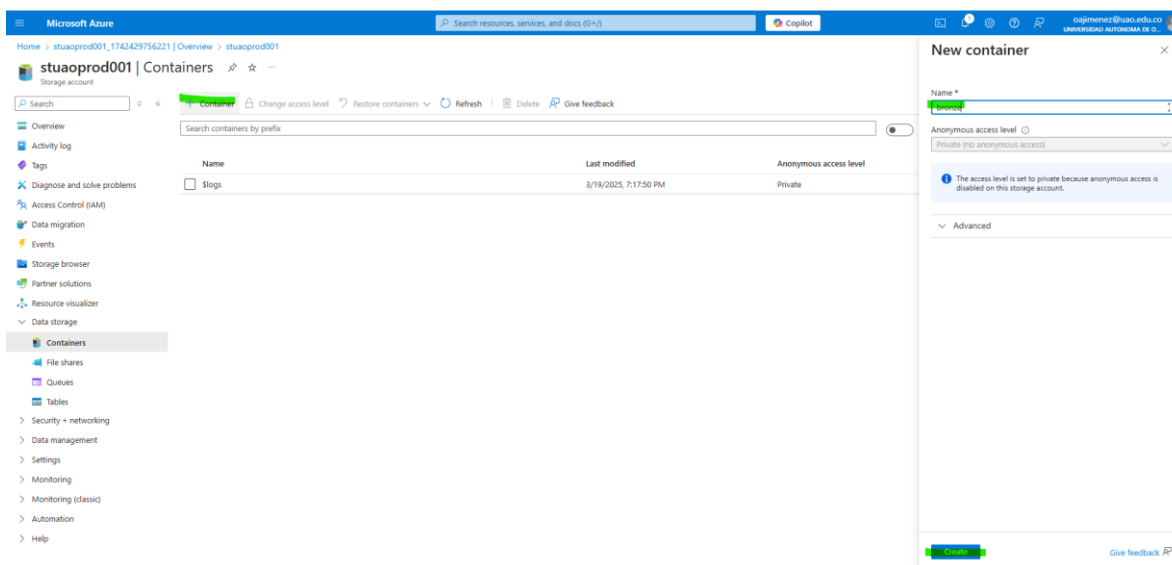
Next steps:

- Go to resource

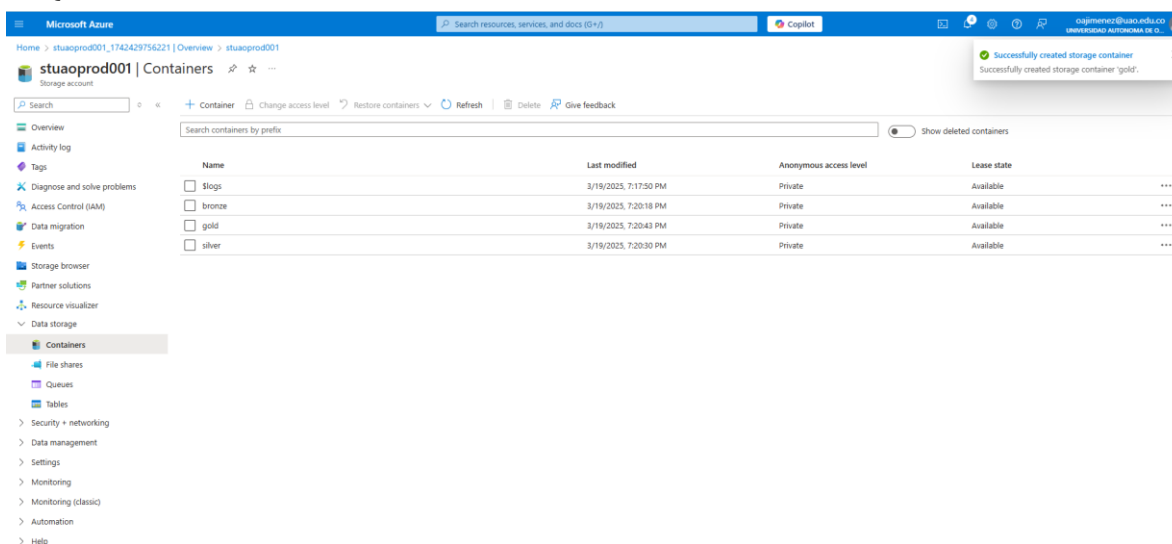
7. Vamos a Almacenamiento de datos y luego en contenedores.



8. Creamos tres contenedores: bronze, silver y gold.



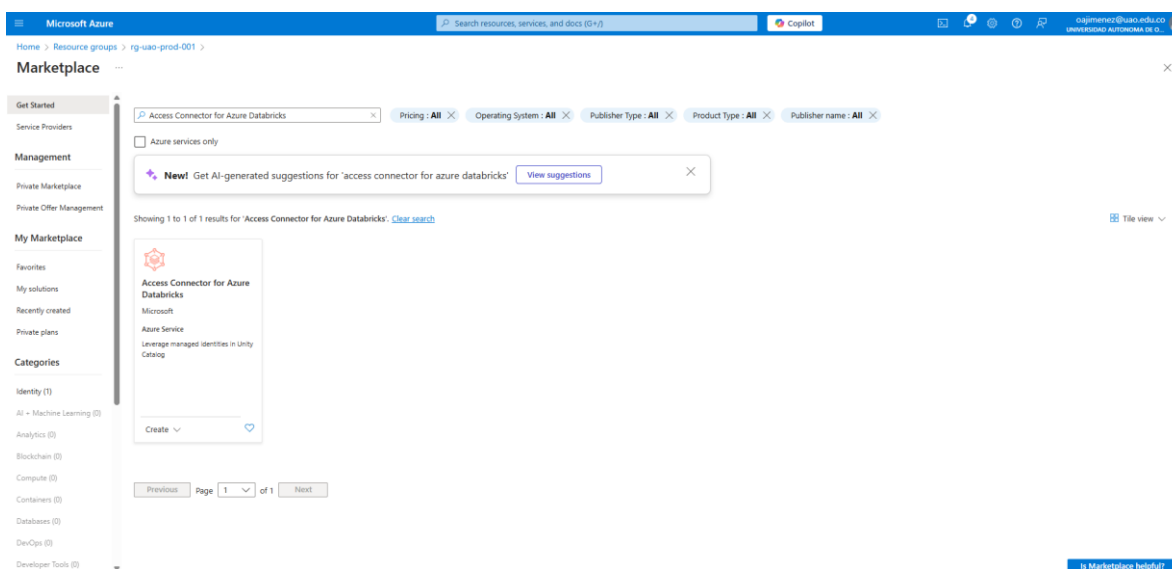
9. Quedarían así:



Name	Last modified	Anonymous access level	Lease state
☐ slogs	3/19/2023, 7:17:50 PM	Private	Available
☐ bronze	3/19/2023, 7:20:18 PM	Private	Available
☐ gold	3/19/2023, 7:20:43 PM	Private	Available
☐ silver	3/19/2023, 7:20:30 PM	Private	Available

Creación de access connector

1. Desde la página del grupo de recursos, damos clic en crear para ingresar a la pestaña del Marketplace, buscamos el Access Connector for Azure Databricks y damos clic en crear.



2. Definimos los siguientes parámetros:

- Grupo de recursos: Creado previamente.
- Nombre: accessconnector-uao-australiacentral
- Región: Australia Central

Hacemos clic en siguiente hasta el final.

The screenshot shows the 'Create an Access Connector for Azure Databricks' wizard in the Microsoft Azure portal. The 'Basics' tab is selected, and the 'Project details' section is visible. The 'Subscription' is set to 'Azure for Students', and the 'Resource group' is 'rg-uao-prod-001'. The 'Instance details' section shows the 'Name' as 'accessconnector-uao-australiacentral' and the 'Region' as 'Australia Central'. The 'Review + create' button is highlighted in blue.

Microsoft Azure

Search resources, services, and docs (0/1)

Copilot

Home > Resource groups > rg-uao-prod-001 > Marketplace >

Create an Access Connector for Azure Databricks

Basics Tags Managed Identity Review + create

The Azure Databricks Access Connector lets you connect managed identities to an Azure Databricks account for the purpose of accessing data registered in Unity Catalog.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource group * rg-uao-prod-001

[Create new](#)

Instance details

Name * accessconnector-uao-australiacentral

Region * Australia Central

Previous Next Review + create

[Give feedback](#)

3. Damos clic en crear

The screenshot shows the 'Create an Access Connector for Azure Databricks' wizard in the Microsoft Azure portal, now at the 'Review + create' tab. The 'Basics' section summarizes the configuration: Subscription (Azure for Students), Resource group (rg-uao-prod-001), Name (accessconnector-uao-australiacentral), and Region (Australia Central). The 'Managed Identity' section shows the identity type as 'SystemAssigned'. The 'Create' button is highlighted in blue.

Microsoft Azure

Search resources, services, and docs (0/1)

Copilot

Home > Resource groups > rg-uao-prod-001 > Marketplace >

Create an Access Connector for Azure Databricks

Basics Tags Managed Identity Review + create

[View automation template](#)

Basics

Subscription Azure for Students

Resource group rg-uao-prod-001

Name accessconnector-uao-australiacentral

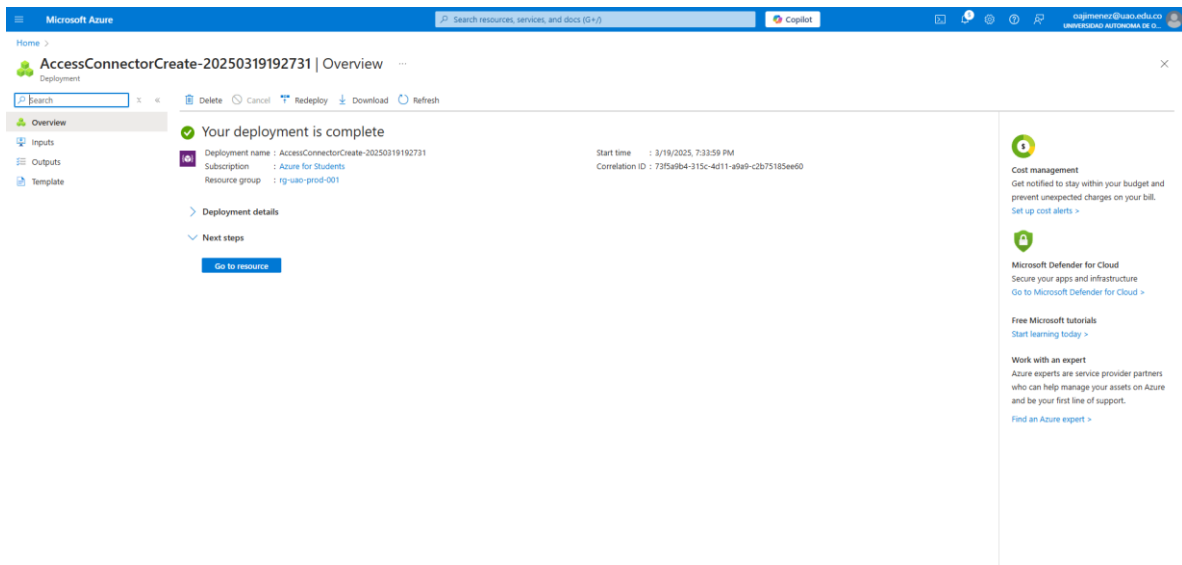
Region Australia Central

Managed Identity

Identity type SystemAssigned

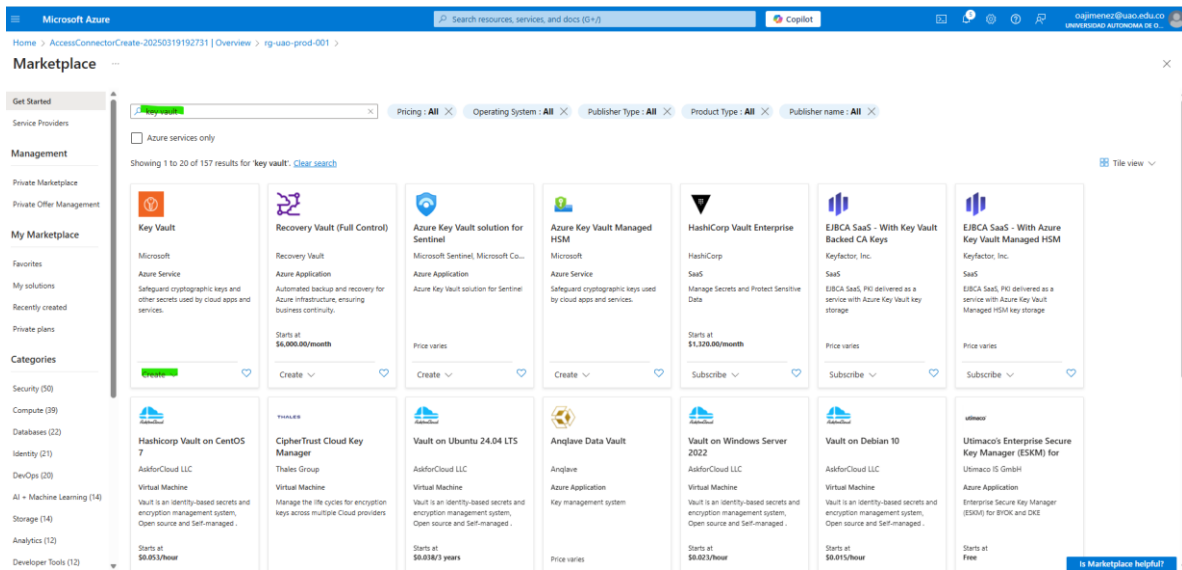
Previous Next Create

[Give feedback](#)



Creación de Key vault

1. Desde la página del grupo de recursos, damos clic en crear para ingresar a la pestaña del Marketplace, buscamos el key vault y damos clic en crear.



2. Definimos los siguientes parámetros:

- Grupo de recursos: El creado previamente.
- Nombre del key vault: kvuao
- Región: Australia Central.

- Tarifa: Standard
- Días de retención de vaults eliminados: 30

Damos clic en siguiente

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Instance details

Key vault name *

Region *

Pricing tier *

Recovery options

Soft delete protection will automatically be enabled on this key vault. This feature allows you to recover or permanently delete a key vault and secrets for the duration of the retention period. This protection applies to the key vault and the secrets stored within the key vault.

To enforce a mandatory retention period and prevent the permanent deletion of key vaults or secrets prior to the retention period elapsing, you can turn on purge protection. When purge protection is enabled, secrets cannot be purged by users or by Microsoft.

Soft-delete ☐ Enabled

Days to retain deleted vaults *

Purge protection ☐ ☒ Disable purge protection (allow key vault and objects to be purged during retention period).

☐ Enable purge protection (enforce a mandatory retention period for deleted vaults and secrets).

[Previous](#) [Next](#) [Review + create](#) [Give feedback](#)

3. Seleccionamos vault acces policy

Access configuration

Configure data plane access for this key vault

To access a key vault in data plane, all callers (users or applications) must have proper authentication and authorization. Authentication establishes the identity of the caller. Authorization determines which operations the caller can execute. [Learn more](#)

Permission model

Grant data plane access by using an Azure RBAC or Key Vault access policy

☐ Azure role-based access control (recommended)

☒ **Key Vault access policy**

Resource access

☐ Azure Virtual Machines for deployment

☐ Azure Resource Manager for template deployment

☐ Azure Disk Encryption for volume encryption

Access policies

Access policies enable you to have fine grained control over access to vault items. [Learn more](#)

[+ Create](#) [Edit](#) [Delete](#)

Name	Email	Key Permissions	Secret Permissions	Certificate Permissions
USER				
☐ Oscar Andres Jimenez Villanueva	oajimenez@uao.edu.co	Get, List, Update, Create, Import, Delete, Recover, Backup, Re...	Get, List, Set, Delete, Recover, Backup, Restore	Get, List, Update, Create, Import, Delete, Recover, Backup, Re...

[Previous](#) [Next](#) [Review + create](#) [Give feedback](#)

4. Damos clic en crear y seleccionamos todos los permisos de secrets

The screenshot shows the 'Create an access policy' dialog in the Microsoft Azure portal. The 'Permissions' tab is selected, and the 'Secret Management Operations' section has all its permissions checked: Select all, Get, List, Set, Delete, Recover, Backup, and Restore. The 'Key Management Operations' and 'Certificate Management Operations' sections are not selected. The 'Principal' tab is also visible, showing the user 'Oscar Andres Jimenez Villanueva' with email 'ojjimenez@uao.edu.co' and permissions 'Get, List, Update, Create, Import, Delete, Recover'.

5. Luego en siguiente buscamos AzureDatabricks para darle permiso a la aplicación.

The screenshot shows the 'Create an access policy' dialog in the Microsoft Azure portal. The 'Principal' tab is selected, and 'AzureDatabricks' is chosen as the principal. The 'Permissions' tab is also visible, showing the same permissions as in the previous screenshot. The 'Principal' tab shows the text 'Only 1 principal can be assigned per access policy. Use the new embedded experience to select a principal. The previous popup experience can be accessed here. Select a principal'.

6. Damos clic en siguiente hasta crear

Microsoft Azure

Home > AccessConnectorCreate-20250319192731 | Overview > rg-uao-prod-001 > Marketplace >

Create a key vault

Basics Access configuration Networking Tags Review + create

Configure data plane access for this key vault

To access a key vault in data plane, all callers (users or applications) must have proper authentication and authorization. Authentication establishes the identity of the caller. Authorization determines which operations the caller can execute. [Learn more](#)

Permission model

Grant data plane access by using a [Azure RBAC](#) or [Key Vault access policy](#)

☐ Azure role-based access control (recommended)

☒ Vault access policy

Resource access

☐ Azure Virtual Machines for deployment

☐ Azure Resource Manager for template deployment

☐ Azure Disk Encryption for volume encryption

Access policies

Access policies enable you to have fine grained control over access to vault items. [Learn more](#)

+ Create

Name	Email	Key Permissions
USER		
Oscar Andres Jimenez Villanueva	oajimenez@uao.edu.co	Get, List, Update, Create, Import, Delete, Recover

Previous Next Review + create

Create an access policy

Permissions Principal Application (optional) Review + create

Key Permissions

Key Management Operations None selected

Cryptographic Operations None selected

Privileged Key Operations None selected

Rotation Policy Operations None selected

Secret Permissions

Secret Management Operations All selected

Privileged Secret Operations None selected

Certificate Permissions

Certificate Management Operations None selected

Privileged Certificate Operations None selected

Principal

Principal name AzureDataBricks

Object ID 717cd945-c360-44c7-a923-a1a5c068b967

Application

Authorized application None selected

Object ID None selected

Previous Create

7. Ya quedó definido el permiso para la aplicación.

Microsoft Azure

Home > AccessConnectorCreate-20250319192731 | Overview > rg-uao-prod-001 > Marketplace >

Create a key vault

Basics Access configuration Networking Tags Review + create

Configure data plane access for this key vault

To access a key vault in data plane, all callers (users or applications) must have proper authentication and authorization. Authentication establishes the identity of the caller. Authorization determines which operations the caller can execute. [Learn more](#)

Permission model

Grant data plane access by using a [Azure RBAC](#) or [Key Vault access policy](#)

☐ Azure role-based access control (recommended)

☒ Vault access policy

Resource access

☐ Azure Virtual Machines for deployment

☐ Azure Resource Manager for template deployment

☐ Azure Disk Encryption for volume encryption

Access policies

Access policies enable you to have fine grained control over access to vault items. [Learn more](#)

+ Create

Name	Email	Key Permissions	Secret Permissions	Certificate Permissions
APPLICATION				
AzureDataBricks			Get, List, Set, Delete, Recover, Backup, Restore	
USER				
Oscar Andres Jimenez Villanueva	oajimenez@uao.edu.co	Get, List, Update, Create, Import, Delete, Recover, Backup, Re...	Get, List, Set, Delete, Recover, Backup, Restore	Get, List, Update, Create, Import, Delete, Recover, Backup, Re...

Previous Next Review + create

[Give feedback](#)

8. Para darle permiso a un usuario, volver a da clic en crear y seleccionamos todos los permisos.

The screenshot shows the 'Create an access policy' dialog in the Microsoft Azure portal. The 'Permissions' tab is selected, and all permissions are checked. The 'Principal' tab is also visible, showing the user 'Oscar Andres Jimenez Villanueva'.

Key permissions	Secret permissions	Certificate permissions
☐ Select all	☒ Select all	☐ Select all
☐ Get	☒ Get	☐ Get
☐ List	☒ List	☐ List
☐ Update	☒ Set	☐ Update
☐ Create	☒ Delete	☐ Create
☐ Import	☒ Recover	☐ Import
☐ Delete	☒ Backup	☐ Delete
☐ Recover	☒ Restore	☐ Recover
☐ Backup		☐ Backup
☐ Restore		☐ Restore
	☐ Privileged Secret Operations	☐ Manage Contacts
	☐ Select all	☐ Manage Certificate Authorities
	☐ Purge	☐ Get Certificate Authorities
		☐ List Certificate Authorities
		☐ Set Certificate Authorities
		☐ Delete Certificate Authorities
		☐ Privileged Certificate Operations

9. Luego en Principal agregamos nuestro propio usuario.

The screenshot shows the 'Create an access policy' dialog in the Microsoft Azure portal. The 'Principal' tab is selected, and the user 'Oscar Andres Jimenez Villanueva' is selected. The 'Permissions' tab is also visible, showing the user 'Oscar Andres Jimenez Villanueva'.

Principal
Oscar Andres Jimenez Villanueva ojjimenez@uao.edu.co

10. Damos clic en siguiente hasta el final y hacemos clic en crear.

Microsoft Azure

Home > AccessConnector/Create-20250319192731 | Overview > rg-uao-prod-001 > Marketplace >

Create a key vault

Configure data plane access for this key vault

To access a key vault in data plane, all callers (users or applications) must have proper authentication and authorization. Authentication establishes the identity of the caller. Authorization determines which operations the caller can execute. [Learn more](#)

Permission model

Grant data plane access by using a [Azure RBAC](#) or [Key Vault access policy](#)

☐ Azure role-based access control (recommended)

☒ Vault access policy

Resource access

☐ Azure Virtual Machines for deployment

☐ Azure Resource Manager for template deployment

☐ Azure Disk Encryption for volume encryption

Access policies

Access policies enable you to have fine grained control over access to vault items. [Learn more](#)

[+ Create](#) [Edit](#) [Delete](#)

Name ↑	Email ↑	Key Permissions
APPLICATION		
☐ AzureDataBricks		
USER		
☐ Oscar Andres Jimenez Villanueva	oajimenez@uao.edu.co	Get, List, Update, Create, Import, Delete, Recover

[Previous](#) [Next](#) [Review + create](#)

Create an access policy

Permissions Principal Application (optional) **Review + create**

Key Permissions

Key Management Operations	None selected
Cryptographic Operations	None selected
Privileged Key Operations	None selected
Rotation Policy Operations	None selected

Secret Permissions

Secret Management Operations	All selected
Privileged Secret Operations	None selected

Certificate Permissions

Certificate Management Operations	None selected
Privileged Certificate Operations	None selected

Principal

Principal name	Oscar Andres Jimenez Villanueva
Object ID	7d7c005a-ef15-43d8-8cd2-4ac56a3a1af

Application

Authorized application	None selected
Object ID	None selected

[Previous](#) [Create](#)

11. Damos clic en siguiente hasta el final, ya que se deja por defecto todo lo demás.

Microsoft Azure

Home > AccessConnector/Create-20250319192731 | Overview > rg-uao-prod-001 > Marketplace >

Create a key vault

Configure data plane access for this key vault

To access a key vault in data plane, all callers (users or applications) must have proper authentication and authorization. Authentication establishes the identity of the caller. Authorization determines which operations the caller can execute. [Learn more](#)

Permission model

Grant data plane access by using a [Azure RBAC](#) or [Key Vault access policy](#)

☐ Azure role-based access control (recommended)

☒ Vault access policy

Resource access

☐ Azure Virtual Machines for deployment

☐ Azure Resource Manager for template deployment

☐ Azure Disk Encryption for volume encryption

Access policies

Access policies enable you to have fine grained control over access to vault items. [Learn more](#)

[+ Create](#) [Edit](#) [Delete](#)

Name ↑	Email ↑	Key Permissions	Secret Permissions	Certificate Permissions
APPLICATION				
☐ AzureDataBricks				
USER				
☐ Oscar Andres Jimenez Villanueva	oajimenez@uao.edu.co	Get, List, Update, Create, Import, Delete, Recover, Backup, Restore	Get, List, Set, Delete, Recover, Backup, Restore	Get, List, Update, Create, Import, Delete, Recover, Backup, Restore
☐ Oscar Andres Jimenez Villanueva	oajimenez@uao.edu.co	Get, List, Update, Create, Import, Delete, Recover, Backup, Restore	Get, List, Set, Delete, Recover, Backup, Restore	Get, List, Update, Create, Import, Delete, Recover, Backup, Restore

[Previous](#) [Next](#) [Review + create](#)

[Give feedback](#)

12. Clic en crear.

The screenshot shows the 'Create a key vault' wizard in the Microsoft Azure portal, specifically the 'Review + create' step. The breadcrumb trail at the top indicates the path: Home > AccessConnector/Create-20250319192731 | Overview > rg-uao-prod-001 > Marketplace > Create a key vault. The wizard has four tabs: Basics, Access configuration, Networking, and Tags, with 'Review + create' being the active tab. A link 'View Automation Template' is present. The configuration is divided into three sections: Basics, Access configuration, and Networking. The Basics section lists: Subscription (Azure for Students), Resource group (rg-uao-prod-001), Key vault name (kvuao), Region (Australia Central), Pricing tier (Standard), Soft-delete (Enabled), Purge protection during retention period (Disabled), and Days to retain deleted vaults (30 days). The Access configuration section lists: Azure Virtual Machines for deployment (Disabled), Azure Resource Manager for template deployment (Disabled), Azure Disk Encryption for volume encryption (Disabled), Permission model (Vault access policy), and Access policies (3). The Networking section lists: Connectivity method (Public endpoint (all networks)). At the bottom, there are 'Previous', 'Next', and 'Create' buttons, with 'Create' being highlighted in blue. A 'Give feedback' link is also visible.

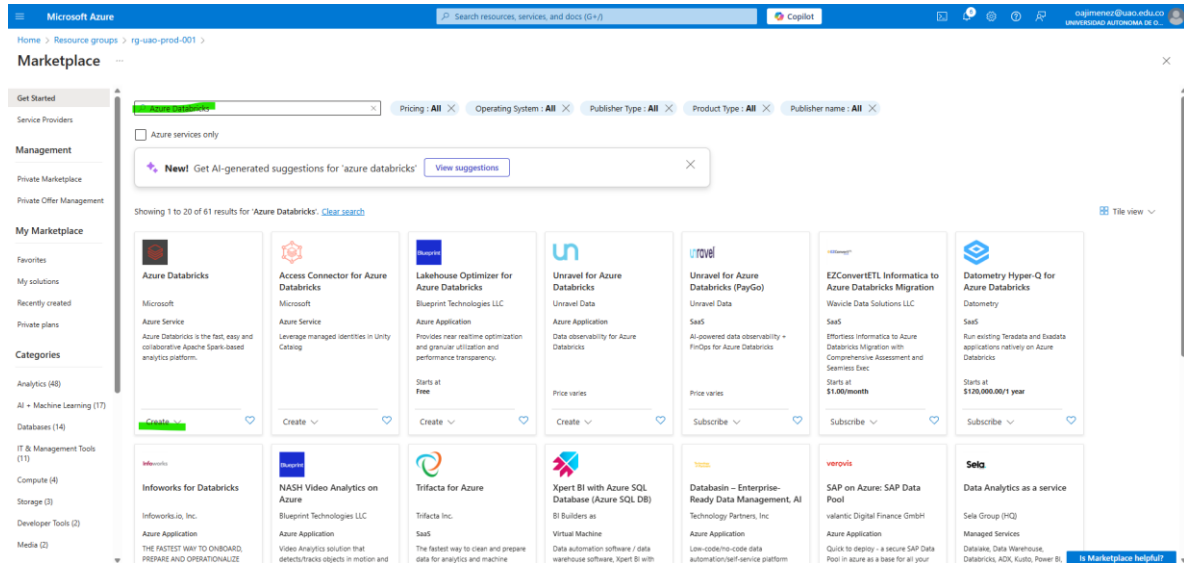
Section	Property	Value
Basics	Subscription	Azure for Students
	Resource group	rg-uao-prod-001
	Key vault name	kvuao
	Region	Australia Central
	Pricing tier	Standard
	Soft-delete	Enabled
	Purge protection during retention period	Disabled
Access configuration	Days to retain deleted vaults	30 days
	Azure Virtual Machines for deployment	Disabled
	Azure Resource Manager for template deployment	Disabled
	Azure Disk Encryption for volume encryption	Disabled
	Permission model	Vault access policy
Access policies		3
	Networking	
Connectivity method		Public endpoint (all networks)

13

The screenshot shows the 'kvuao | Overview' page in the Microsoft Azure portal. The breadcrumb trail is: Home > kvuao | Overview. The page has a search bar and a toolbar with 'Delete', 'Cancel', 'Redeploy', 'Download', and 'Refresh' buttons. The left sidebar shows 'Overview' as the selected tab, with other options being 'Inputs', 'Outputs', and 'Template'. The main content area displays a green checkmark and the message 'Your deployment is complete'. Below this, the deployment details are shown: Deployment name: kvuao, Subscription: Azure for Students, and Resource group: rg-uao-prod-001. The start time is 3/19/2025, 8:01:48 PM, and the correlation ID is 25b8339c-b610-49d0-828e-c59b4695c99. There are links for 'Deployment details', 'Next steps', and a 'Go to resource' button. A 'Give feedback' link is also present. The right sidebar contains three sections: 'Cost management' (Get notified to stay within your budget and prevent unexpected charges on your bill. Set up cost alerts >), 'Microsoft Defender for Cloud' (Secure your apps and infrastructure. Go to Microsoft Defender for Cloud >), and 'Free Microsoft tutorials' (Start learning today >). At the bottom, there is a 'Work with an expert' section (Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support. Find an Azure expert >).

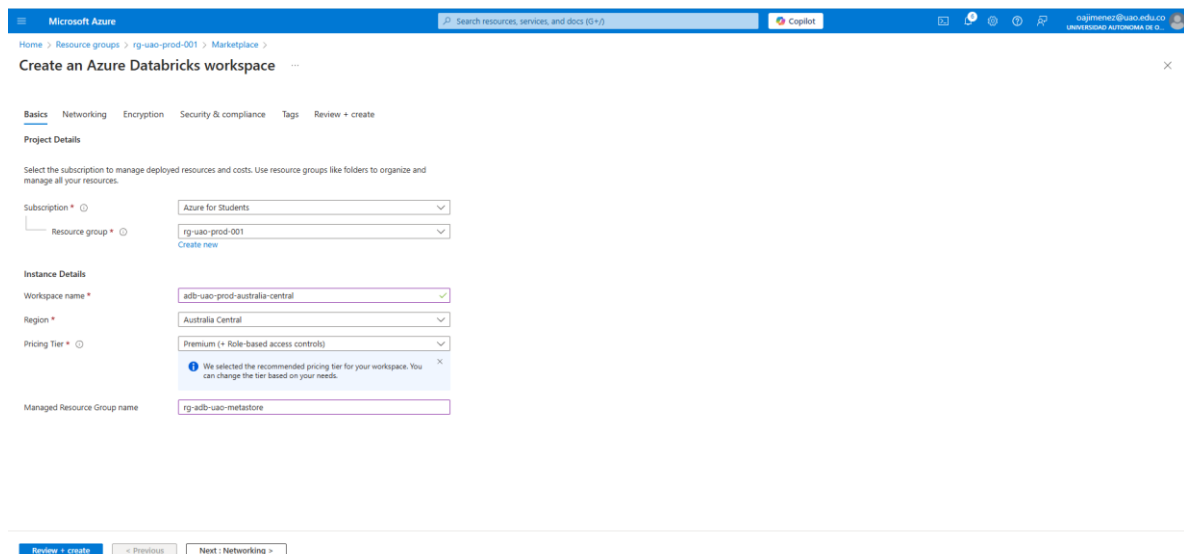
Creación de espacio de trabajo de Databricks

1. Desde la página del grupo de recursos, damos clic en crear para ingresar a la pestaña del Marketplace, buscamos por Azure Databricks y damos clic en crear.



2. Definimos los siguientes parámetros:

- Grupo de recursos: El creado previamente rg-uao-prod-001.
- Nombre del espacio de trabajo: adb-uao-prod-australia-central.
- Región: Australia Central.
- Tarifa: Premium.
- Grupo de recursos administrado: rg-adb-uao-metastore



3. Hacemos clic en siguiente hasta revisar y crear

The screenshot shows the 'Create an Azure Databricks workspace' wizard in the Microsoft Azure portal. The 'Tags' tab is selected, showing a table with columns 'Name', 'Value', and 'Resource'. The 'Resource' column is populated with 'Azure Databricks Service'. At the bottom, there are buttons for 'Review + create' and '< Previous'.

Name	Value	Resource
		Azure Databricks Service

4. Hacemos clic en crear

The screenshot shows the 'Review + create' screen of the 'Create an Azure Databricks workspace' wizard. A green banner at the top indicates 'Validation Succeeded'. The screen displays configuration details for the workspace, including Basics, Networking, Encryption, and Security & compliance settings.

Validation Succeeded

Basics

Workspace name	adb-ua0-prod-australia-central
Subscription	Azure for Students
Resource group	rg-ua0-prod-001
Region	Australia Central
Pricing Tier	premium
Managed Resource Group name	rg-adb-ua0-metastore

Networking

Deploy Azure Databricks workspace with Secure Cluster Connectivity (for Public IP)	No
Deploy Azure Databricks workspace in your own Virtual Network (vnet)	No

Encryption

Enable Infrastructure Encryption	No
Enable CMK for Managed Disks	No
Enable CMK for Managed Services	No

Security & compliance

Compliance Security Profile	No
Compliance Standards	No
Enhanced Security Monitoring	No
Automatic Cluster Update	No

At the bottom, there are buttons for 'Create', '< Previous', and a link to 'Download a template for automation'.

5

Microsoft Azure

Search resources, services, and docs (0/1)

Exploit

rg-uao-prod-001_adb-uao-prod-australia-central | Overview

Home

rg-uao-prod-001_adb-uao-prod-australia-central | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : rg-uao-prod-001_adb-uao-prod-australia-central

Subscription : Azure for Students

Resource group : rg-uao-prod-001

Start time : 3/16/2025, 8:12:08 PM

Correlation ID : 3886892-5440-40fb-ad1d-f779d528fa2a

> Deployment details

< Next steps

Go to resource

Give feedback

Tell us about your experience with deployment

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure

Go to Microsoft Defender for Cloud >

Free Microsoft tutorials

Start learning today >

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.

Find an Azure expert >

Ya tenemos nuestros 4 recursos creados:

Microsoft Azure

Search resources, services, and docs (Ctrl+K)

Copilot

Home >

rg-uao-prod-001

Resource group

Search

- Create
- Manage view
- Delete resource group
- Refresh
- Export to CSV
- Open query
- Assign tags
- Move
- Delete
- Export template
- Open in mobile

Essentials

Subscription ([manage](#)) : [Azure for Students](#)

Subscriptions ID : 1d73b3bb-92f7-4c4e-a85c-0ea6cb15c15

Tags ([edit](#)) : [Add tags](#)

Deployments : [4 Succeeded](#)

Location : Australia Central

Resources

Recommendations

Filter for any field... Type equals all Location equals all Add filter

Showing 1 to 4 of 4 records. Show hidden types

Name ↑↓	Type ↑↓	Location ↑↓
accessconnector-uao-australiacentral	Access Connector for Azure Databricks	Australia Central
adb-uao-prod-australia-central	Azure Databricks Service	Australia Central
kvuao	Key vault	Australia Central
stuakigprod001	Storage account	Australia Central

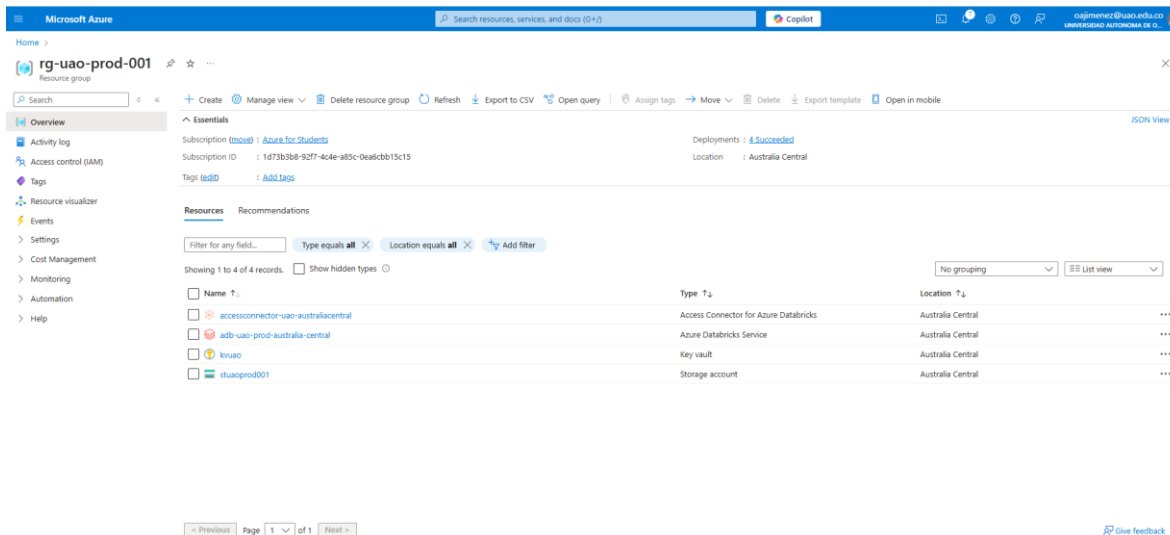
< Previous Page 1 of 1 Next >

Give feedback

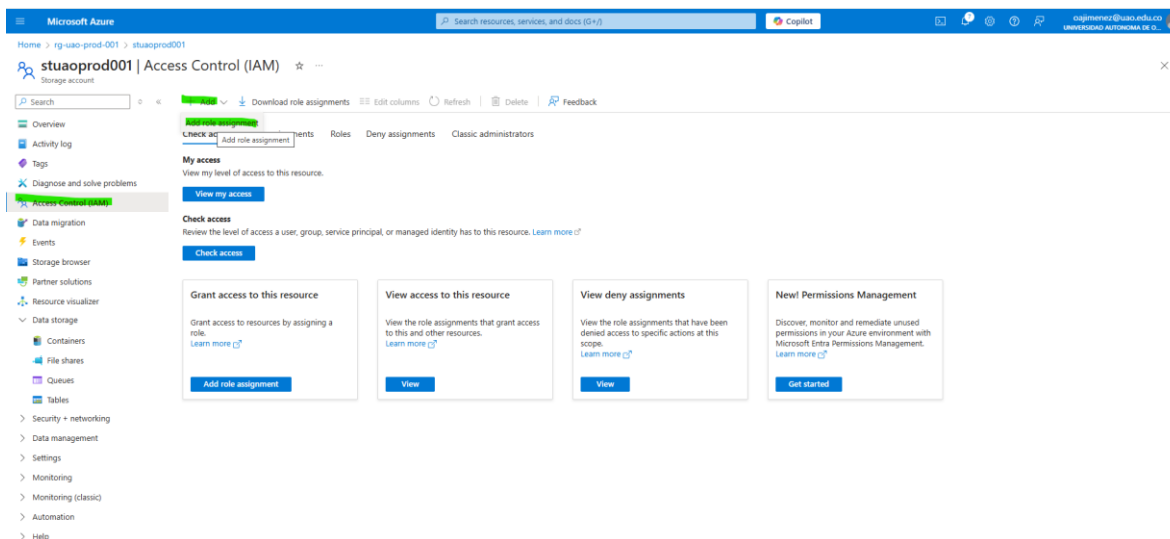
Guía elaborada por: Prof. Oscar Andrés Jiménez Villanueva.

Permisos de conexión a la cuenta de almacenamiento desde el Access connector de Databricks

1 . Ingresamos a la cuenta de almacenamiento



2. En la cuenta de almacenamiento dar clic en Access control (IAM) y luego en agregar



3. Buscamos por rol storage blob data contributor y damos clic en siguiente.

Microsoft Azure

Home > rg-uao-prod-001 > shuasprod001 | Access Control (IAM) >

Add role assignment

Role **Members** Conditions Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#) >

Job function roles Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

storage blob data contributor x Type: All Category: All

Name ↑	Description ↑	Type ↑	Category ↑	Details
Storage Blob Data Contributor	Allows for read, write and delete access to Azure Storage blob containers and data	BuiltInRole	Storage	View

Showing 1 - 1 of 1 results.

Review + assign Previous Next

Feedback

4. Seleccionamos managed identity y luego en select members

Microsoft Azure

Home > rg-uao-prod-001 > shuasprod001 | Access Control (IAM) >

Add role assignment

Role **Members** Conditions Review + assign

Selected role Storage Blob Data Contributor

Assign access to ☐ User, group, or service principal ☒ Managed identity

Members

Name	Object ID	Type
No members selected		

Description

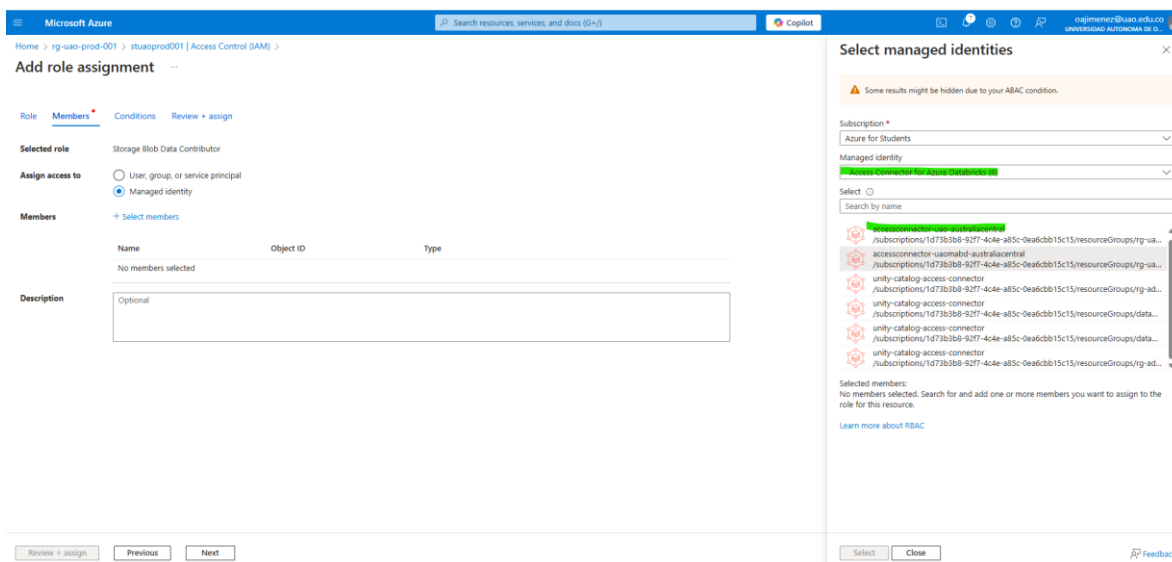
Optional

Review + assign Previous Next

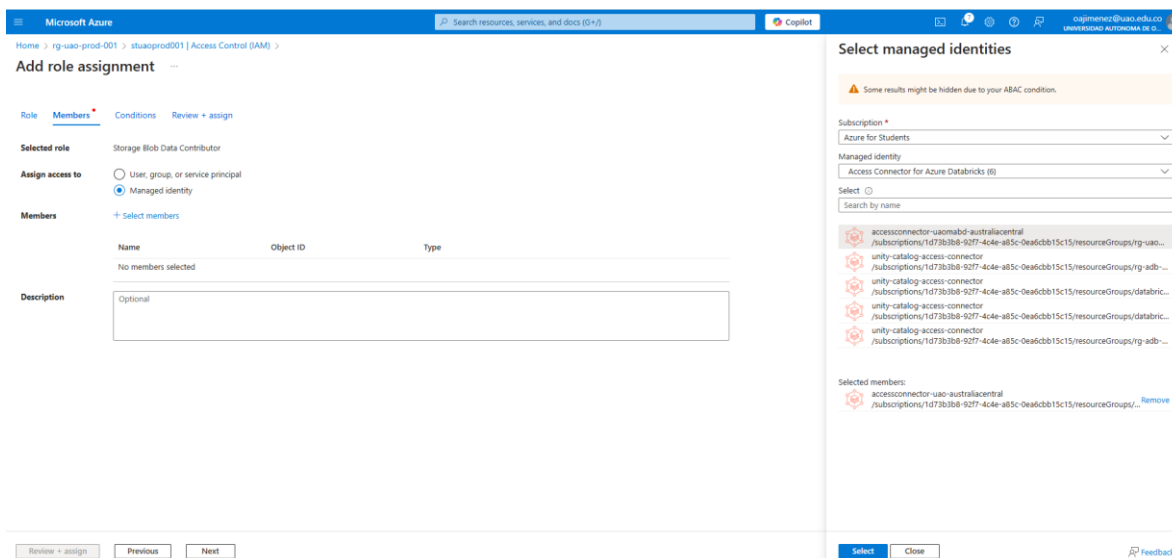
Feedback

5. Seleccionamos los parámetros:

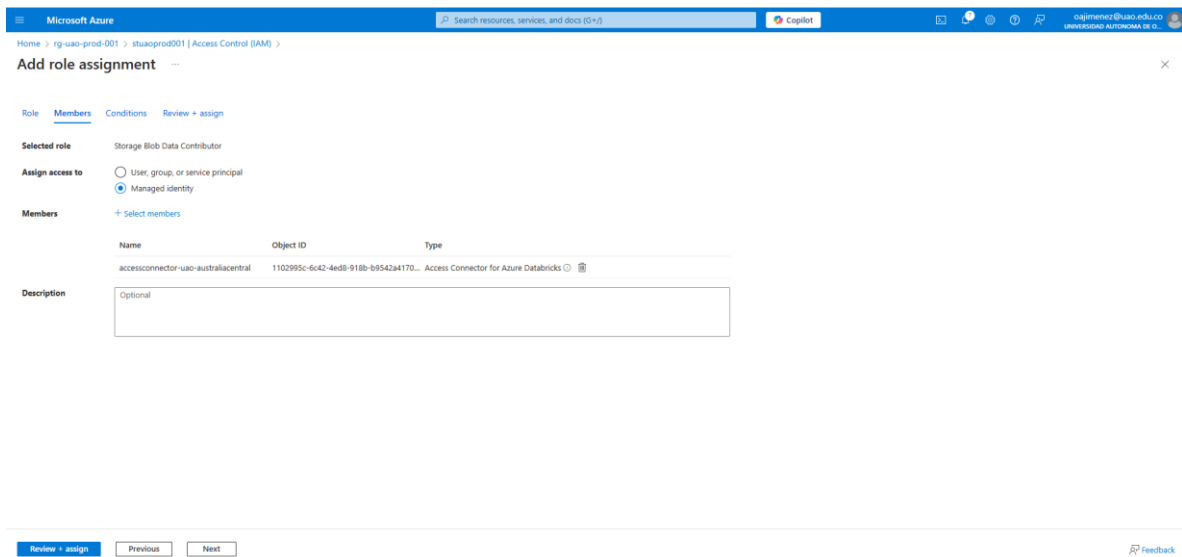
- Manage identity: Azure connector for azure databricks
- Select: El nombre del acces connector creado previamente que es accessconnector-uao-australia-central



6. De nuevo damos clic en select en la parte inferior

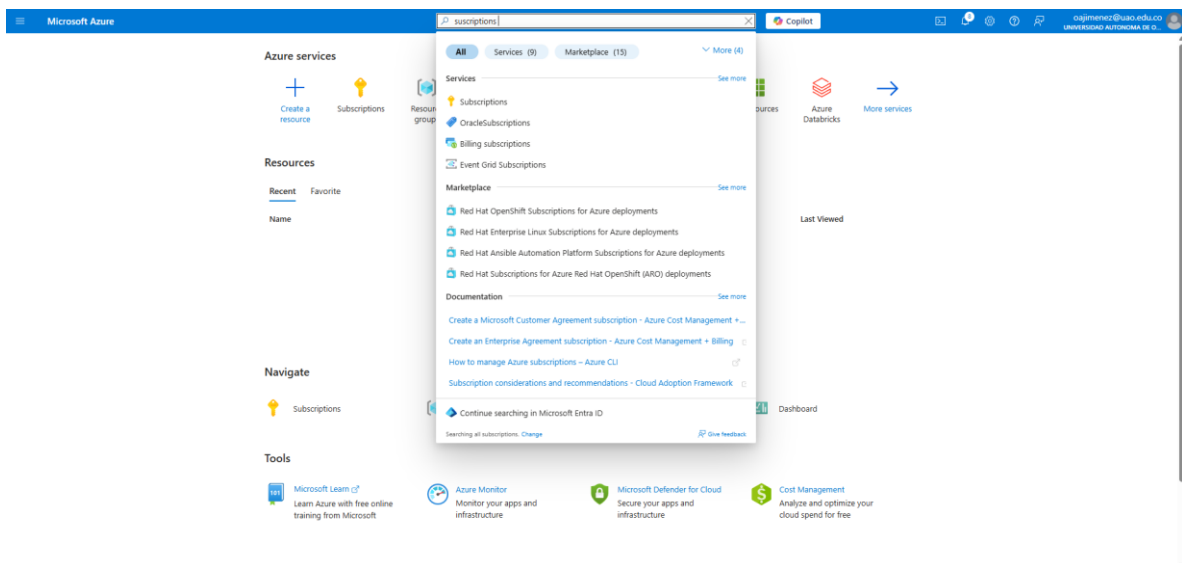


7. Damos clic en review + assign

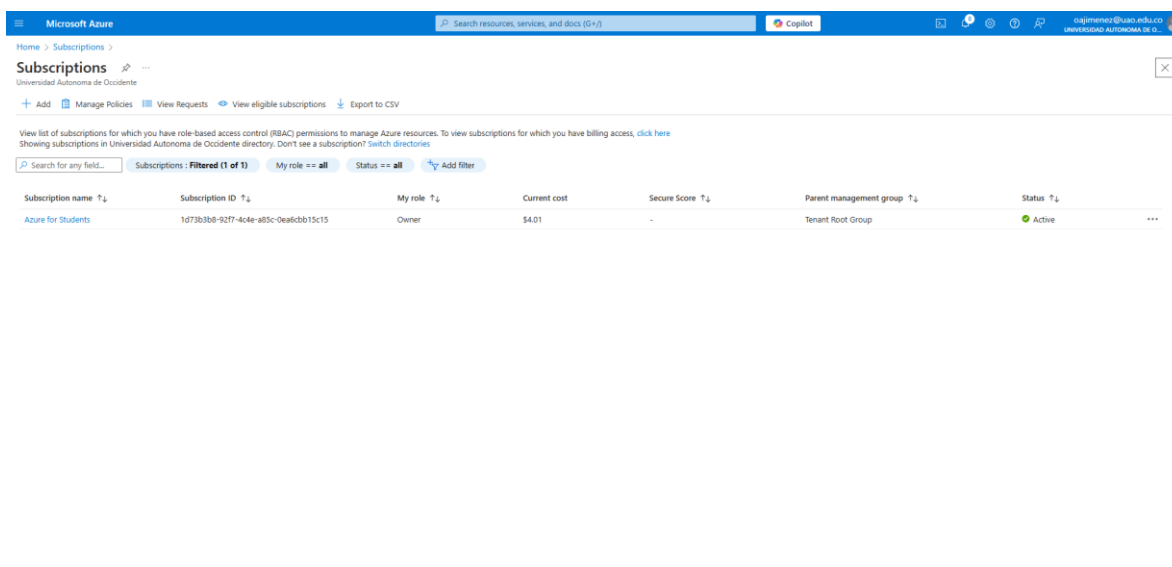


Permisos de usuario sobre la suscripción (opcional)

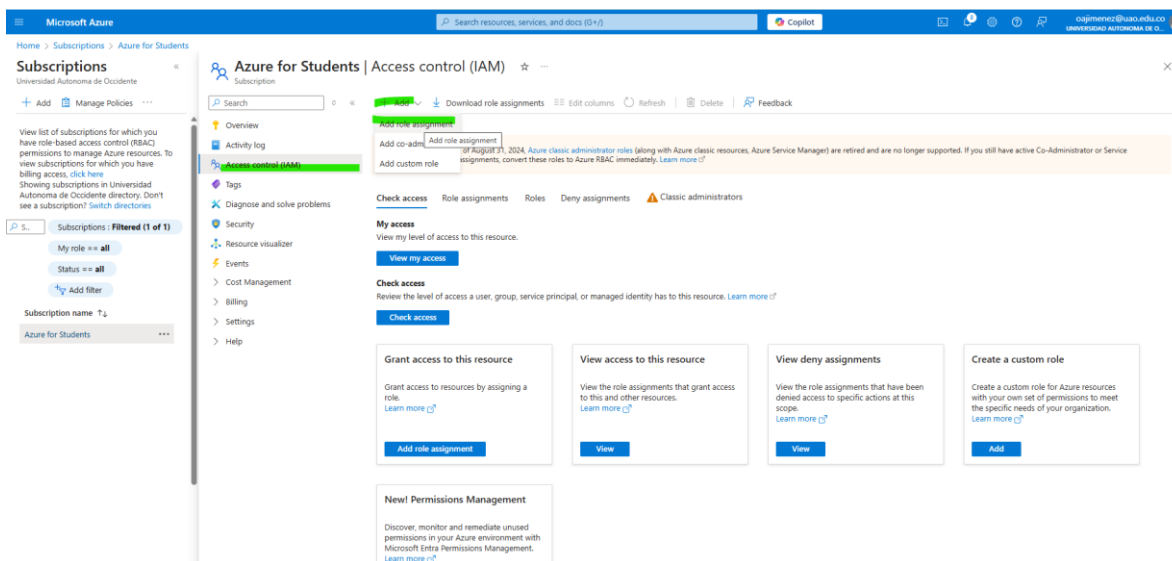
1. Desde home vamos a suscripciones



2. Ingresamos a nuestra suscripción de estudiante



3. Ingresamos a IAM y seleccionamos add role assignment



4. Seleccionamos privileged administrator roles y luego en contributor

Microsoft Azure

Home > Subscriptions > Azure for Students | Access control (IAM)

Add role assignment

Role Members Conditions Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Job function roles **Privileged administrator roles**

Grant privileged administrator access, such as the ability to assign roles to other users.

Can a job function role with less access be used instead?

Name	Description	Type	Category	Details
Owner	Grants full access to manage all resources, including the ability to assign roles in Azure RBAC.	BuiltInRole	General	View
Contributor	Grants full access to manage all resources, but does not allow you to assign roles in Azure RBAC, manage assignments in Azure Blueprints, or share image gal...	BuiltInRole	General	View
Role Based Access Control Administrator	Manage access to Azure resources by assigning roles using Azure RBAC. This role does not allow you to manage access using other ways, such as Azure Policy.	BuiltInRole	None	View
User Access Administrator	Lets you manage user access to Azure resources.	BuiltInRole	General	View

Showing 1 - 4 of 4 results.

[Review + assign](#) [Previous](#) [Next](#)

5. Seleccionar assing acces to, luego en + select members, ingresamos el correo del usuario lo seleccionamos y finalmente en select.

Home > Subscriptions > Azure for Students | Access control (IAM)

Add role assignment

Role **Members** Conditions Review + assign

Selected role Contributor

Assign access to ☒ User groups or service principal ☐ Managed identity

Members

Name	Object ID	Type
No members selected		

Description

Optional

Select members

oajimenez

Oscar Andres Jimenez Villanueva
oajimenez@uao.edu.co

Selected members:

No members selected. Search for and add one or more members you want to assign to the role for this resource.

[Learn more about RBAC](#)

[Select](#) [Close](#)

[Review + assign](#) [Previous](#) [Next](#)

https://portal.azure.com/?Microsoft_Azure_Education_correlationid=3a77c940-81c2-4e38-988-03213e4022863&Microsoft_Azure_Education_new4d2=trun&Microsoft_Azure_Education_aosSubGuid=1d77b3b6-8207-4e4e-ad5c-0ae63bb15c15f

6. Damos clic en review + assign

Home > Subscriptions > Azure for Students | Access control (IAM)

Add role assignment

Role **Members** Conditions Review + assign

Selected role Contributor

Assign access to ☒ User, group, or service principal ☐ Managed identity

Members + Select members

Name	Object ID	Type
Oscar Andres Jimenez Villanueva	7d7c005a-6f15-45c6-8cf2-14c56e2ea1af	User

Description Optional

Review + assign Previous Next

https://portal.azure.com/Microsoft_Azure_Education_correlationId=2677e940-81c3-4c38-99f-05233e322b33&Microsoft_Azure_Education_new4d=trued&Microsoft_Azure_Education_acoSubGuid=1d73b3b8-9207-4c4e-a85c-0ea6c3a15c15f8

7. Vemos que se ha asignado el usuario a la suscripción como contributor.

Microsoft Azure Search resources, services, and docs (G+7) Copilot oajimenez@uaq.edu.co

Home > Subscriptions > Azure for Students

Subscriptions

Universidad Autonoma de Occidente

+ Add Manage Policies

View list of subscriptions for which you have role-based access control (RBAC) permissions to manage Azure resources. To view subscriptions for which you have billing access, click [here](#). Showing subscriptions in Universidad Autonoma de Occidente directory. Don't see a subscription? [Switch directories](#)

Subscriptions: Filtered (1 of 1)

My role == all

Status == all

Add filter

Subscription name ↑

Azure for Students

Azure for Students | Access control (IAM)

Subscription

Search

+ Add Download role assignments Edit columns Refresh Delete Feedback

Overview Activity log Access control (IAM) Tags Diagnose and solve problems Security Resource visualizer Events Cost Management Billing Settings Help

Action required: As of August 31, 2024, Azure classic administrator roles (along with Azure classic resources, Azure Service Manager) are retired and are no longer supported. If you still have active Co-Administrator or Service Administrator role assignments, convert these roles to Azure RBAC immediately. [Learn more](#)

Check access Role assignments Roles Deny assignments Classic administrators

Number of role assignments for this subscription 13 4000 Privileged 3

View assignments

Search by name or email Type: All Role: All Scope: All scopes Group by: Role

All (3) Job function roles (0) Privileged administrator roles (3)

Name	Type	Role	Scope	Condition
Owner (2)				
☐ Oscar Andres Jimenez Villanueva oajimenez@uaq.edu.co	User	Owner	This resource	Add
☐ Oscar Andres Jimenez Villanueva oajimenez@uaq.edu.co	User	Owner	This resource	Add
Contributor (1)				
☐ Oscar Andres Jimenez Villanueva oajimenez@uaq.edu.co	User	Contributor	This resource	None

Showing 1 - 3 of 3 results.

Parte 2

Despliegue cluster de Databricks

1. Desde el grupo de recursos ingresamos a Azure Databricks

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and the user profile. The main content area is titled 'Resource groups' and shows the selected resource group 'rg-uao-prod-001'. The 'Overview' tab is active, displaying the following information:

- Subscription: [\(model\) : Azure for Students](#)
- Location: Australia Central
- Deployments: [4 Succeeded](#)
- Tags: [Add tags](#)

Below this information, there is a table of resources. The table has columns for Name, Type, and Location. The resources listed are:

Name	Type	Location
accessconnector-uao-australiacentral	Access Connector for Azure Databricks	Australia Central
adb-uao-prod-australia-central	Azure Databricks Service	Australia Central
kvuao	Key vault	Australia Central
stuaoprod001	Storage account	Australia Central

2. Damos clic en launch workspace

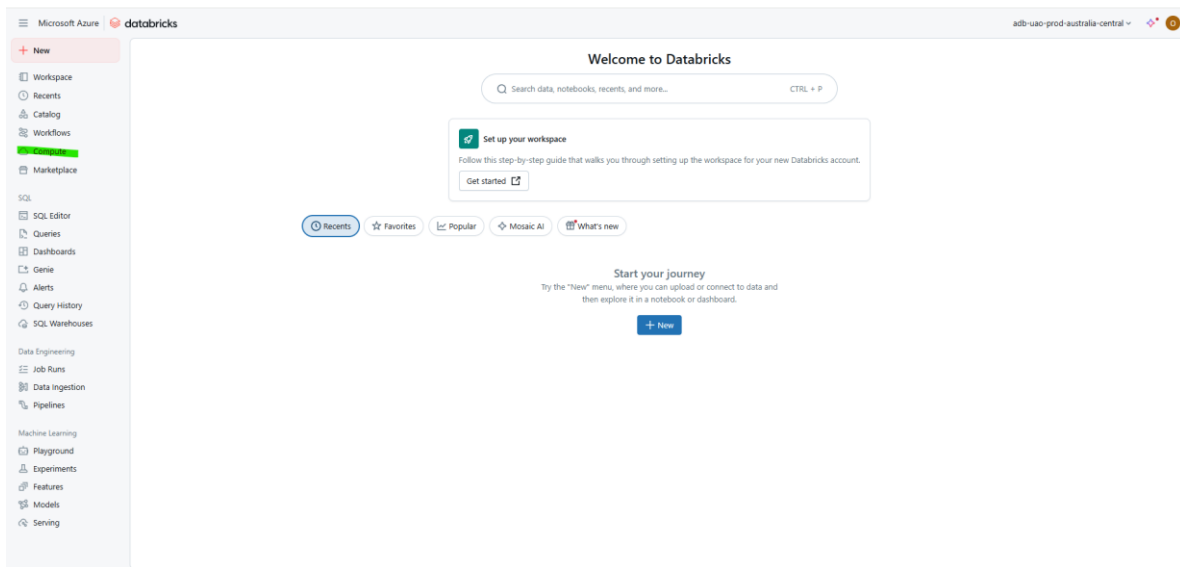
The screenshot shows the Microsoft Azure portal interface for the 'adb-uao-prod-australia-central' resource. The 'Overview' tab is active, displaying the following information:

- Status: Active
- Managed Resource Group: [rg-adb-uao-metastore](#)
- URL: [https://adb-67905053812996.16.azure.databricks.net](#)
- Pricing Tier: [Premium \(1+ Role-based access controls\) \(Click to change\)](#)

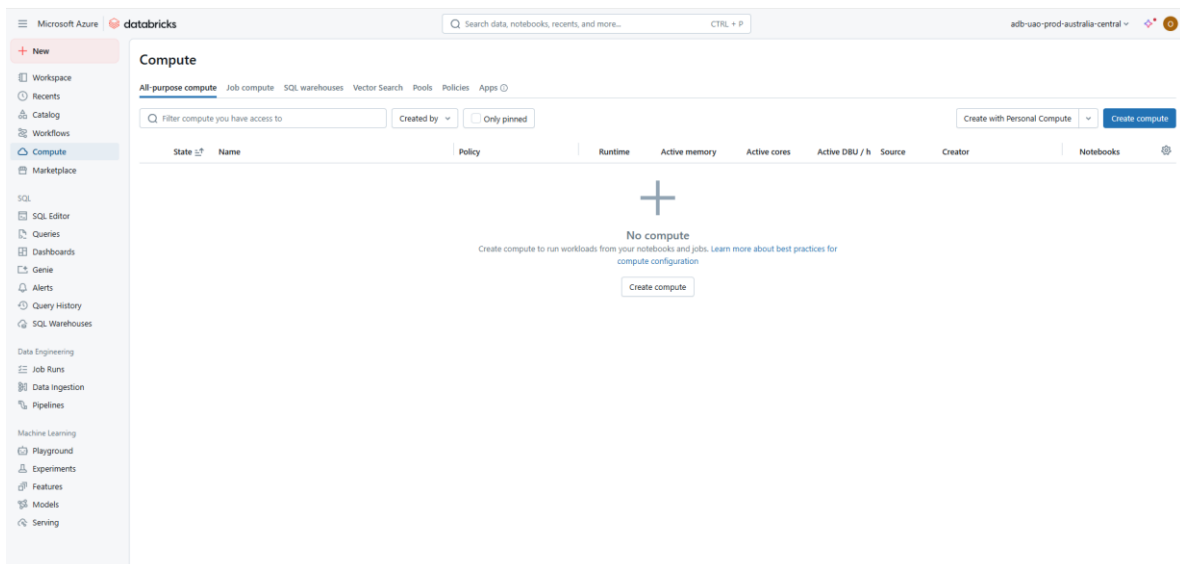
Below this information, there is a 'Launch Workspace' button. Below the button, there is a grid of links for various resources:

Documentation	Getting Started	Import Data from File	Import Data from Azure Storage
Notebook	Admin Guide	Link Azure ML workspace	

3. Hacemos clic en compute



4. Crear computo



5. Definimos los siguientes parametros:

- Nombre del Cluster: Cluster-uao-Prod
- Policy: unrestricted.
- Single node
- Acces mode: Dedicated

- Runtime: Por defecto.
- Photon acceleration: sin marcar.
- Node type: Standard_DS3_v2
- Terminate after: 15 minutos de inactividad.

Microsoft Azure | databricks | Search data, notebooks, recent, and more... | CTRL + P | adb-uao-prod-australia-central

+ New

- Workspace
- Recents
- Catalog
- Workflows
- Compute**
- Marketplace
- SQL
- SQL Editor
- Queries
- Dashboards
- Genie
- Alerts
- Query History
- SQL Warehouses
- Data Engineering
- Job Runs
- Data ingestion
- Pipelines
- Machine Learning
- Playground
- Experiments
- Features
- Models
- Serving

Compute > New compute > Single form: OFF

Cluster-uao-Prod

Policy: Unrestricted

Access mode: Single user access

Dedicated (formerly: Single user) | Oscar Villa...

Performance

Databricks runtime version: Runtime: 15.4 LTS (Scala 2.12, Spark 3.5.0)

Use Photon Acceleration

Node type: Standard_DS3_v2 | 14 GB Memory, 4 Cores

Terminate after: 15 minutes of inactivity

Tags

Add tags

Key | Value | Add

> Automatically added tags

> Advanced options

Create compute | Cancel

Summary

1 Driver | 14 GB Memory, 4 Cores

Runtime: 15.4x-scala2.12

Unity Catalog | Standard_DS3_v2 | 0.75 DBU/h

Damos clic en create compute.

6. Ya queda listo nuestro clúster de computo

Microsoft Azure | databricks | Search data, notebooks, recent, and more... | CTRL + P | adb-uao-prod-australia-central

+ New

- Workspace
- Recents
- Catalog
- Workflows
- Compute**
- Marketplace
- SQL
- SQL Editor
- Queries
- Dashboards
- Genie
- Alerts
- Query History
- SQL Warehouses
- Data Engineering
- Job Runs
- Data ingestion
- Pipelines
- Machine Learning
- Playground
- Experiments
- Features
- Models
- Serving

Compute > Single form: OFF

Cluster-uao-Prod

Configuration | Notebooks (0) | Libraries | Event log | Spark UI | Driver logs | Metrics | Apps | Spark compute UI - Master

Policy: Unrestricted

Access mode: Single user access

Dedicated (formerly: Single user) | Oscar Villanueva

Performance

Databricks Runtime Version: 15.4 LTS (includes Apache Spark 3.5.0, Scala 2.12)

Use Photon Acceleration

Node type: Standard_DS3_v2 | 14 GB Memory, 4 Cores

Terminate after: 15 minutes of inactivity

Tags

No custom tags

> Automatically added tags

> Advanced options

Terminate | Edit

Summary

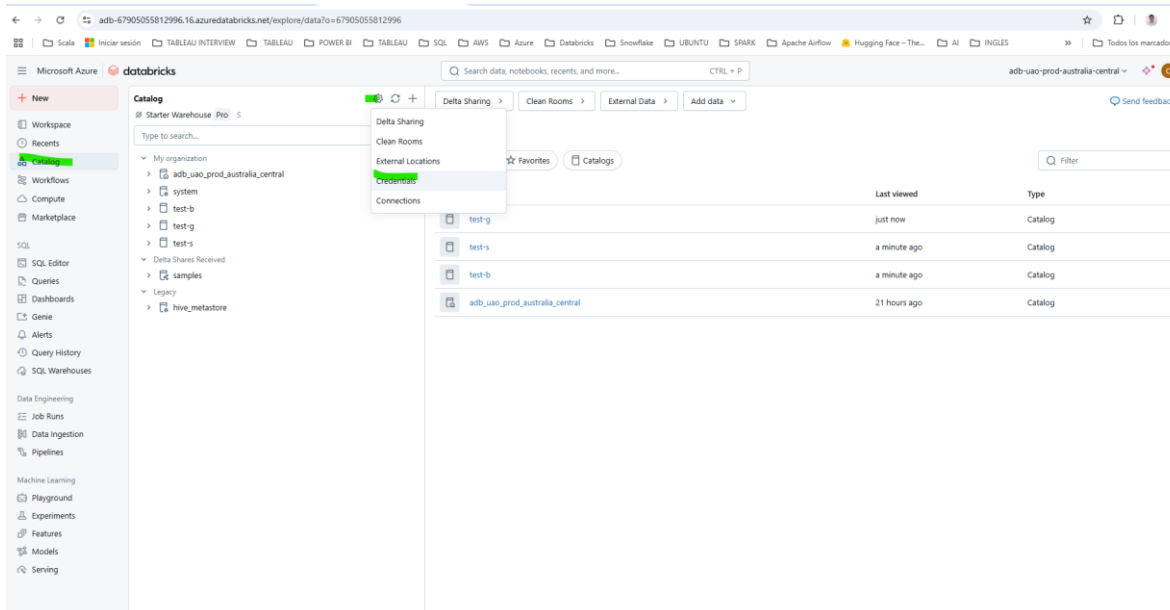
1 Driver | 14 GB Memory, 4 Cores

Runtime: 15.4x-scala2.12

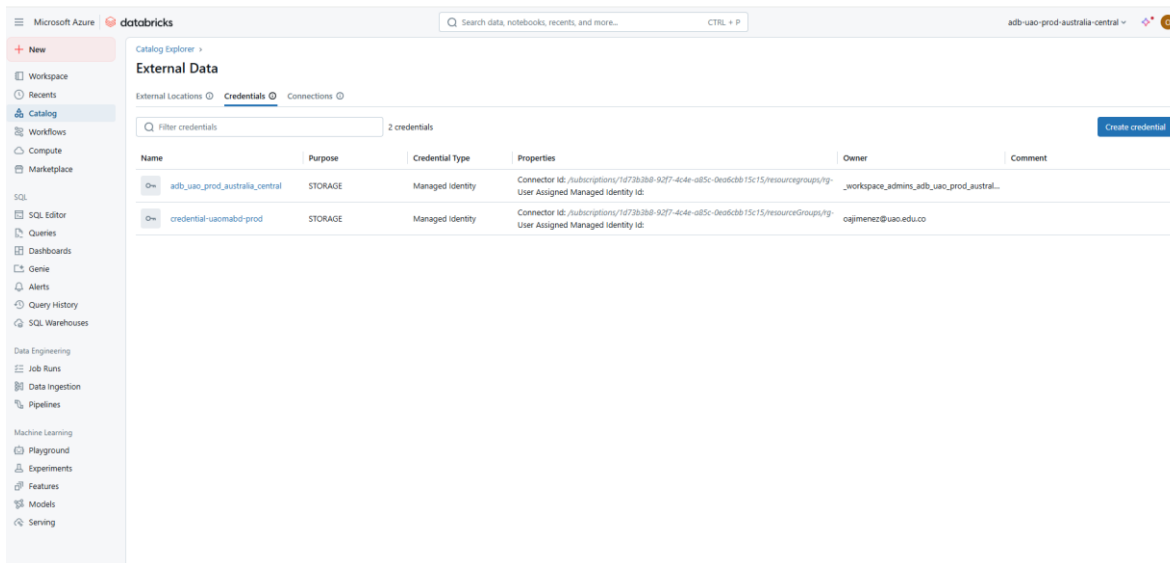
Unity Catalog | Standard_DS3_v2 | 0.75 DBU/h

Crear Credencial de almacenamiento

1. Ingresamos a catalog, luego damos clic en Manage y finalmente en credentials.

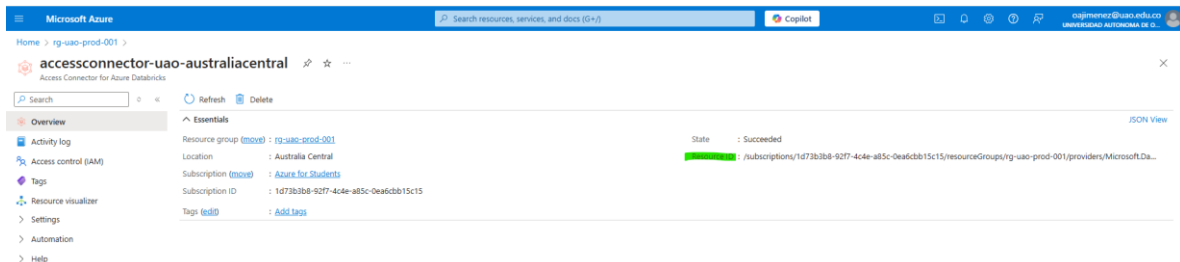


2. Damos clic en create credential

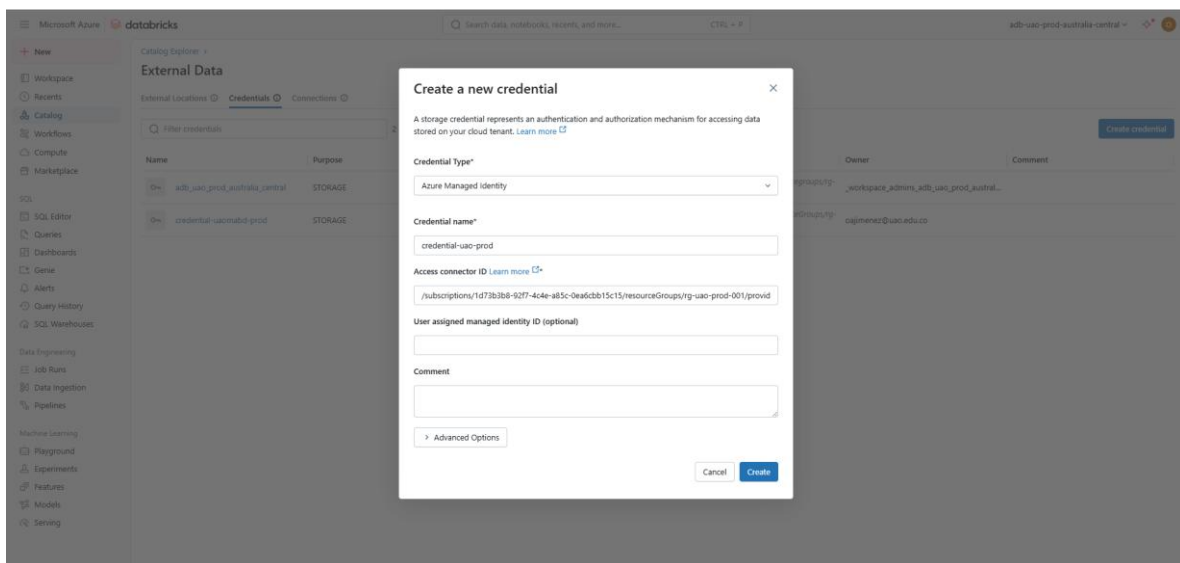


3. Definimos los siguientes parámetros:

- Credential name: credential-uao-prod
- Acces connector ID: Debemos ir al Access connector creado previamente y extraer el resource ID

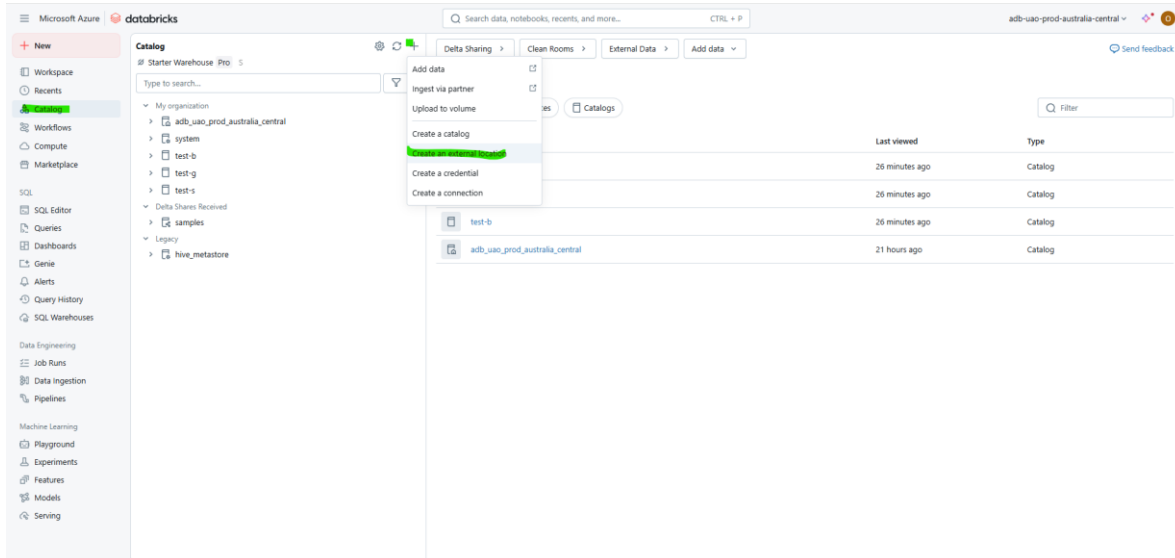


Luego damos clic en crear



Creación de external location

1. Desde la pestaña de Catalog, damos clic en + y luego en create an external location



2. Este paso se debe hacer para los tres catálogos que crearemos, para los contenedores de bronze, silver y gold.

Definimos el nombre como:

- ext-loc-catalog-bronze
- ext-loc-catalog-silver
- ext-loc-catalog-gold

En el campo de URL para cada external location seria:

- abfss://bronze@stuaoprod001.dfs.core.windows.net/
- abfss://silver@stuaoprod001.dfs.core.windows.net/
- abfss://gold@stuaoprod001.dfs.core.windows.net/

Seleccionamos el credential previamente creado y damos crear en create.

Microsoft Azure databricks

Search data, notebooks, recent, and more... CTRL + P

adb-uao-prod-australia-central

+ New

- Workspace
- Recents
- Catalog**
- Workflows
- Compute
- Marketplace

SQL

- SQL Editor
- Queries
- Dashboards
- Genie
- Alerts
- Query History
- SQL Warehouses

Data Engineering

- Job Runs
- Data Ingestion
- Pipelines

Machine Learning

- Playground
- Experiments
- Features
- Models
- Serving

External Locations

Create a new external location

An external location allows you to access your data stored in cloud storage (e.g. ADLS). You will need the cloud storage path and a paired credential (e.g. managed identity) which gives access to that path. [Learn more](#)

External location name*

ext-loc-catalog-bronze

URL*

Enter the bucket path that you want to use as the external location. Note: This must be an ADLS Gen2 storage account with a hierarchical namespace

abfsc://bronze@stuaomabprod.dfs.core.windows.net/ [Copy from DBFS](#)

Storage credential* [Learn more](#)

Provide a storage credential capable of accessing the URL

credential-uao-prod (Storage credential)

Connector id: /subscriptions/1d73b3b8-9277-4c4e-a85c-0ea6bb15c15/resourceGroups/tg-uao-prod-001/providers/Mic...
User Assigned Managed Identity id:

Comment

[Advanced Options](#)

Cancel **Create**

4. Damos clic en test connection

Microsoft Azure databricks

Search data, notebooks, recent, and more... CTRL + P

adb-uao-prod-australia-central

+ New

- Workspace
- Recents
- Catalog**
- Workflows
- Compute
- Marketplace

SQL

- SQL Editor
- Queries
- Dashboards
- Genie
- Alerts
- Query History
- SQL Warehouses

Data Engineering

- Job Runs
- Data Ingestion
- Pipelines

Machine Learning

- Playground
- Experiments
- Features
- Models
- Serving

Catalog Explorer > External Locations >

ext-loc-catalog-bronze

[test connection](#)

Overview Permissions Browse Workspaces

Description

[Add description](#)

Credential	credential-uao-prod
URL	abfsc://bronze@stuaomabprod001.dfs.core.windows.net/
Limit to read-only use	Disabled

About this external location

Owner: Oscar Villanueva

Fallback mode: ☐

Microsoft Azure databricks

Search data, notebooks, recent, and more... CTRL + P

adb-uao-prod-australia-central

+ New

- Workspace
- Recents
- Catalog**
- Workflows
- Compute
- Marketplace

SQL

- SQL Editor
- Queries
- Dashboards
- Genie
- Alerts
- Query History
- SQL Warehouses

Data Engineering

- Job Runs
- Data Ingestion
- Pipelines

Machine Learning

- Playground
- Experiments
- Features
- Models
- Serving

Catalog Explorer > External Locations >

ext-loc-catalog-bronze

[test connection](#)

Overview Permissions Browse Workspaces

Description

[Add description](#)

Credential	credential-uao-prod
URL	abfsc://bronze@stuaomabprod001.dfs.core.windows.net/
Limit to read-only use	Disabled

About this external location

Owner: Oscar Villanueva

Fallback mode: ☐

Test connection

Location Type: Directory

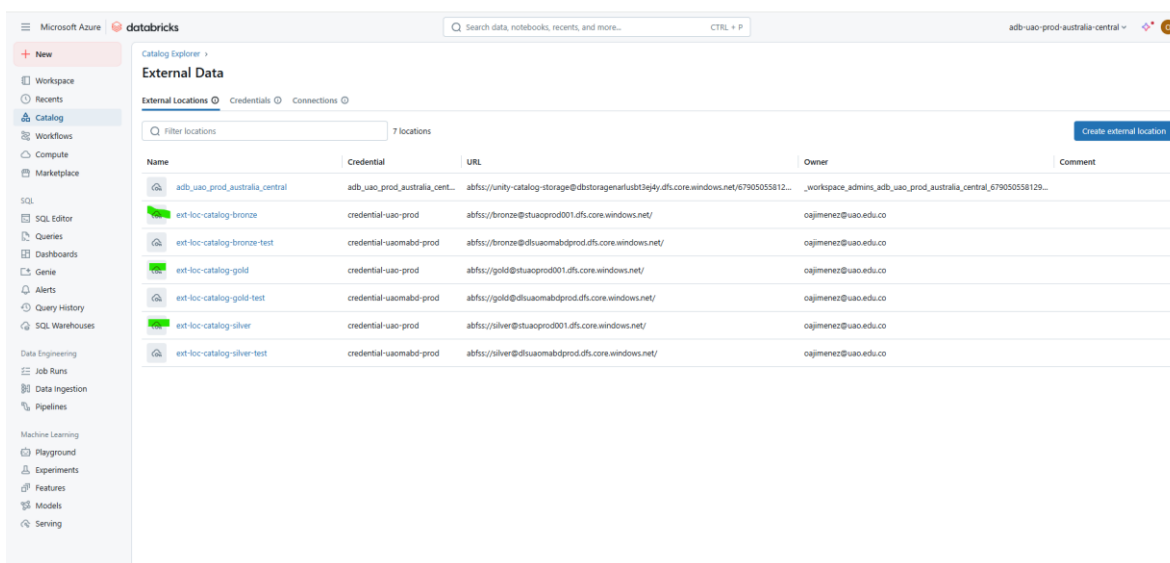
- Success - Read
- Success - List
- Success - Write
- Success - Delete
- Success - Path Exists
- Success - Hierarchical Namespace Enabled

All Permissions Confirmed

The associated Storage Credential grants permission to perform all necessary operations.

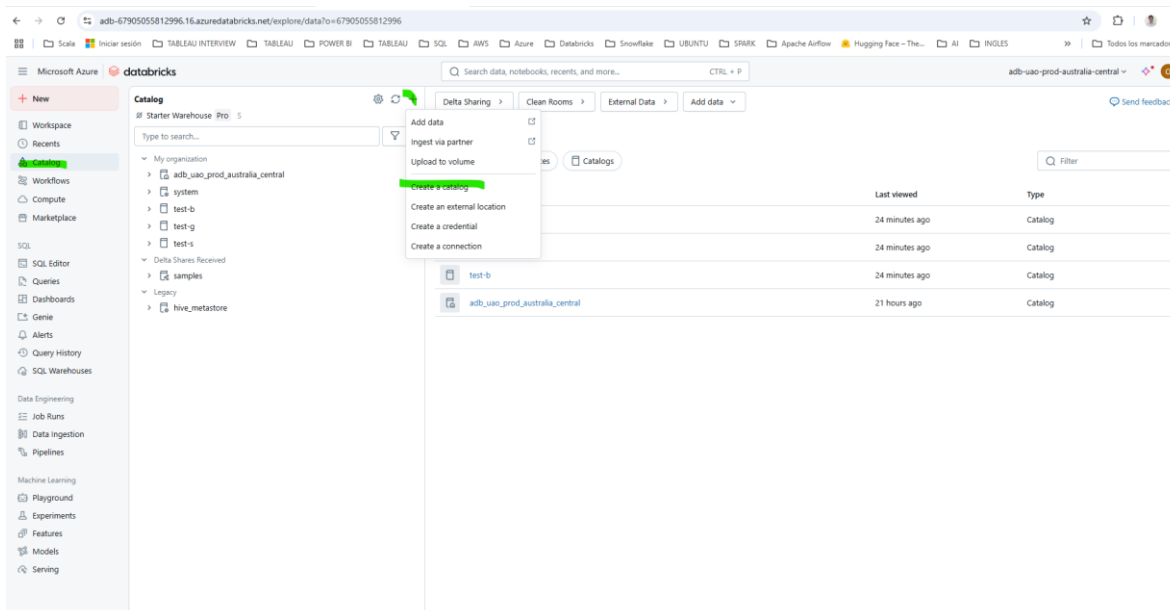
Done

5. Tenemos nuestros 3 external location creados



Creación de Catálogos

1. Desde la pestaña de Catalog, damos clic en + y luego en créate a catalog

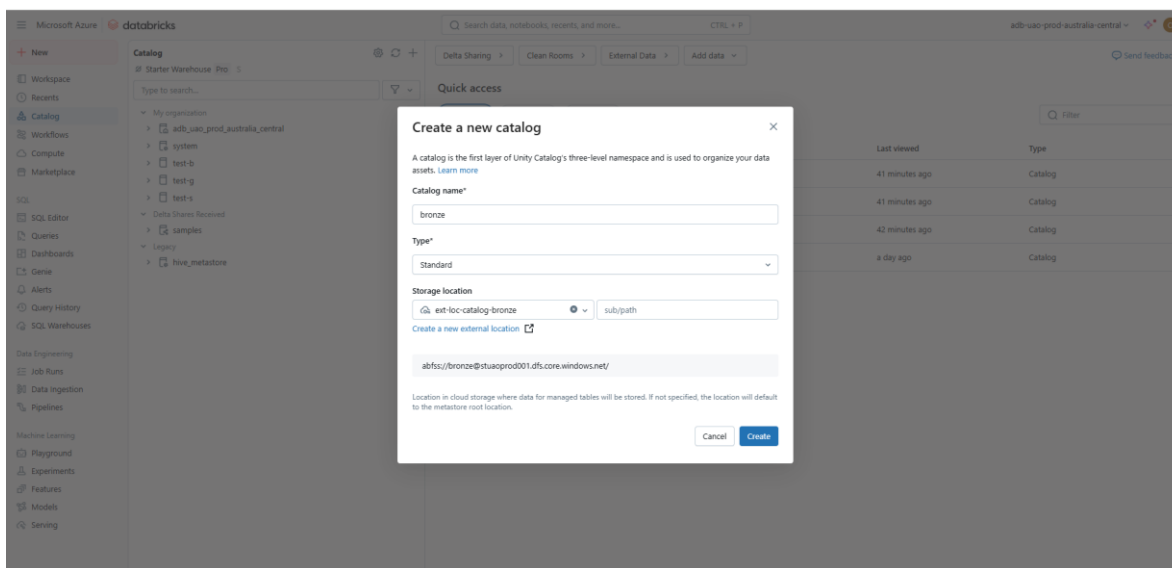


2. Este paso lo debemos repetir tres veces para crear los catálogos bronze, silver y gold.

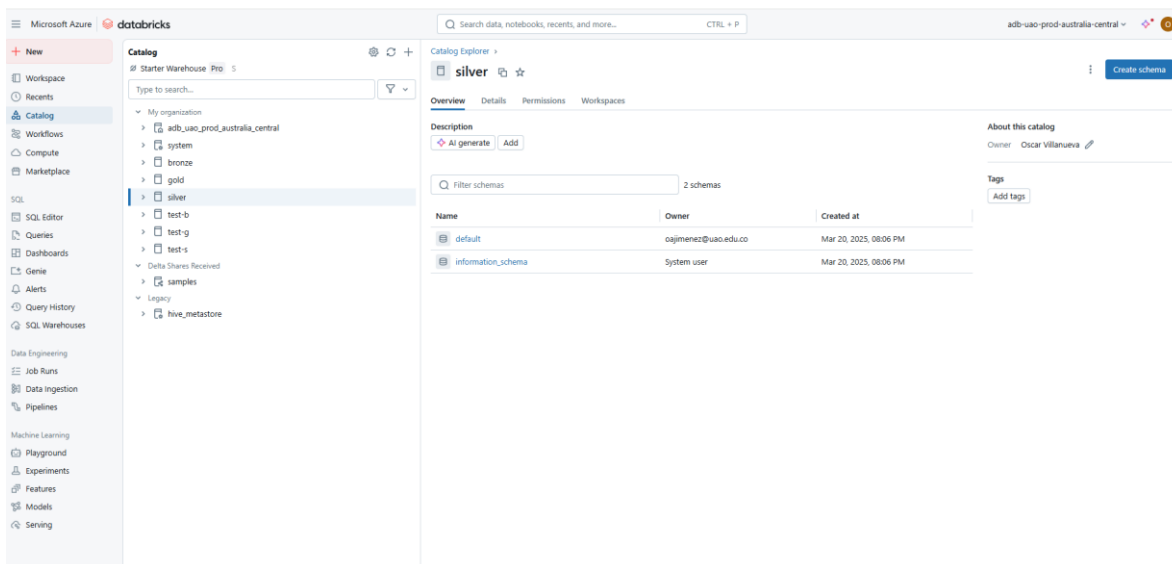
Catalog name: bronze

Storage location: Seleccionamos el external location de bronze

Lo demás se deja como en la imagen y damos clic en crear.



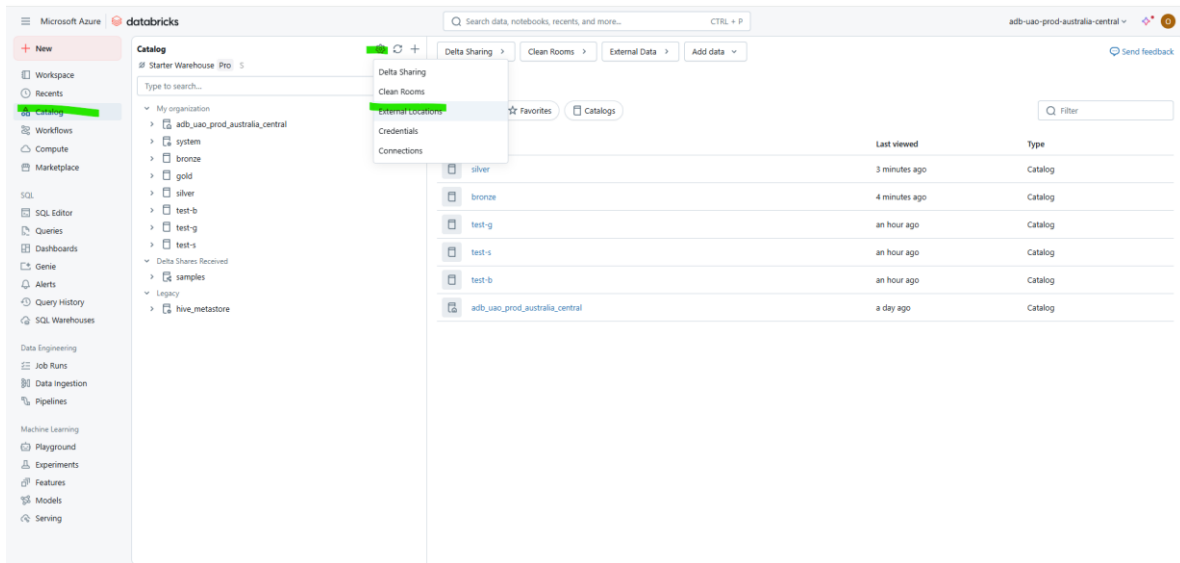
3. Así quedarían nuestros 3 catálogos:



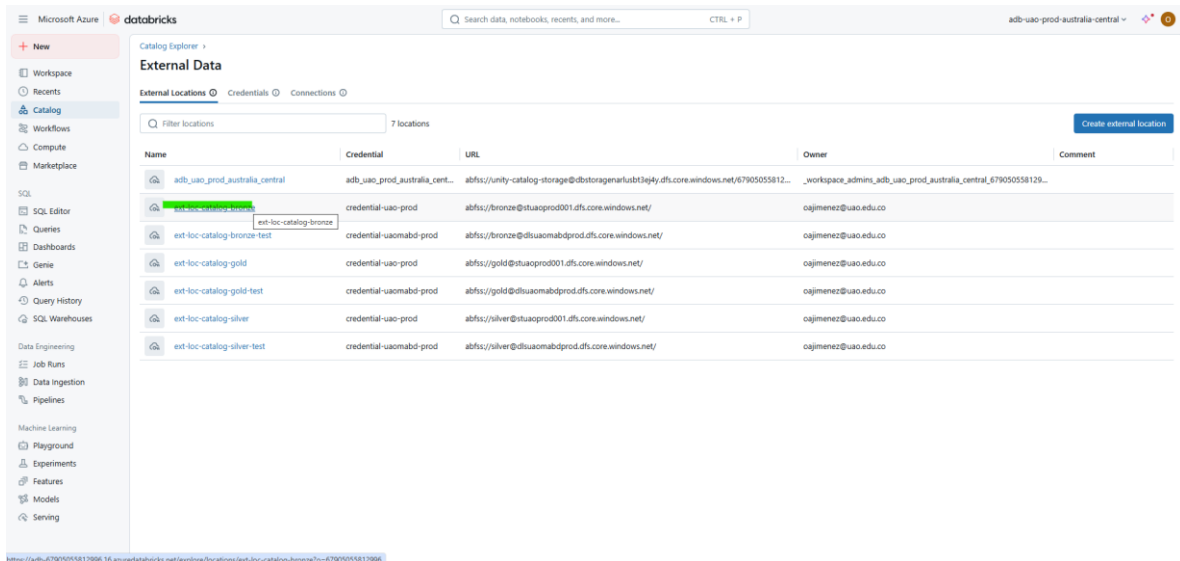
Otorgar permisos a los external location

Esto se debe hacer por cada uno de los 3 external location creados previamente.

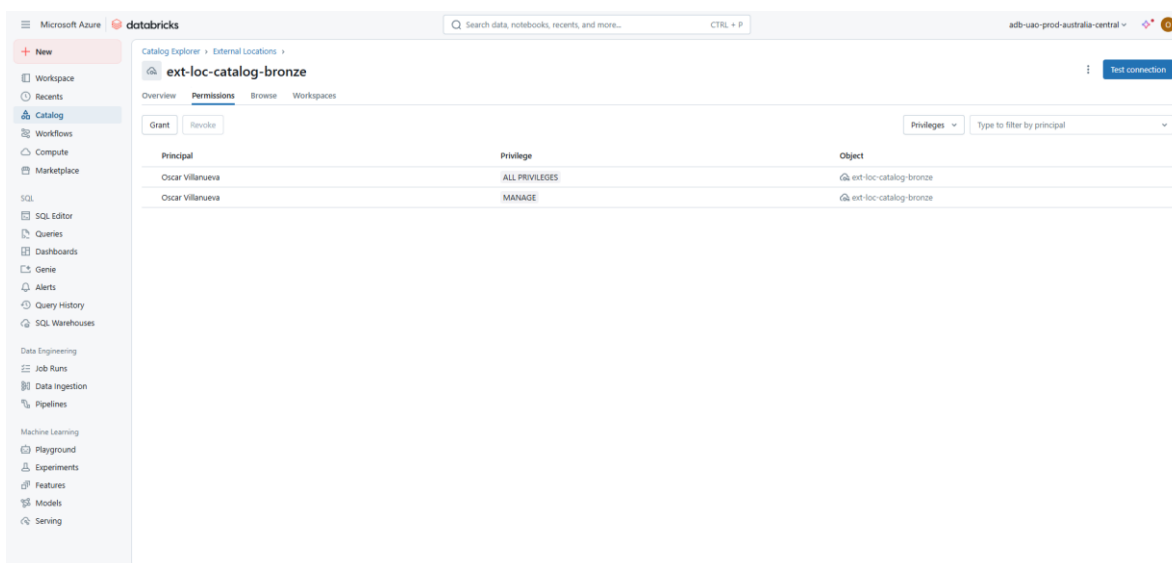
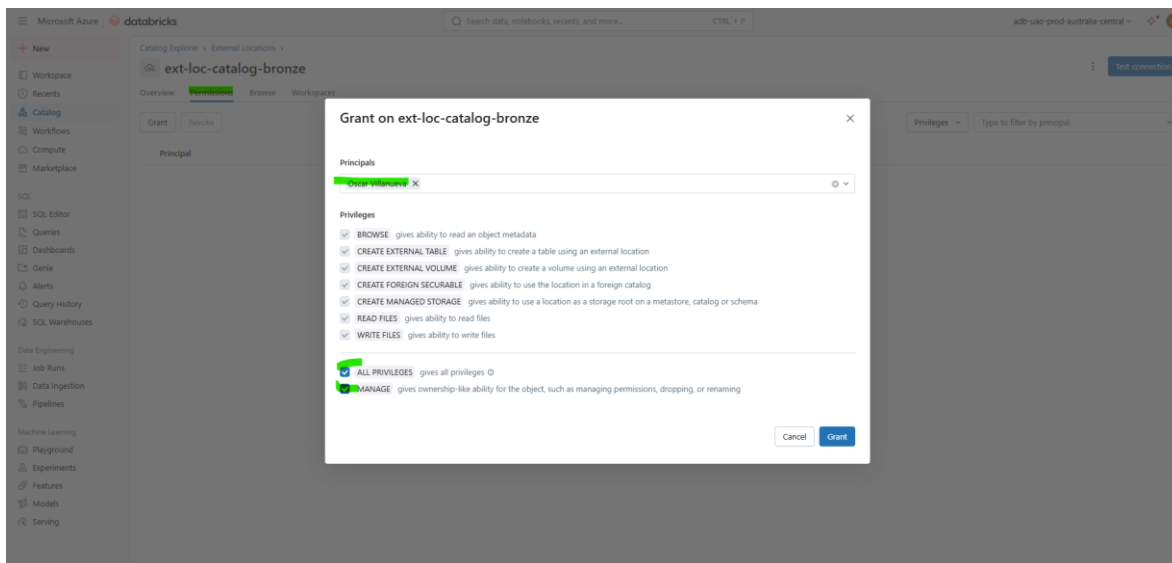
1. Desde la pestaña de Catalog, damos clic en Manage y luego en create an external location



2. Seleccionamos el external location



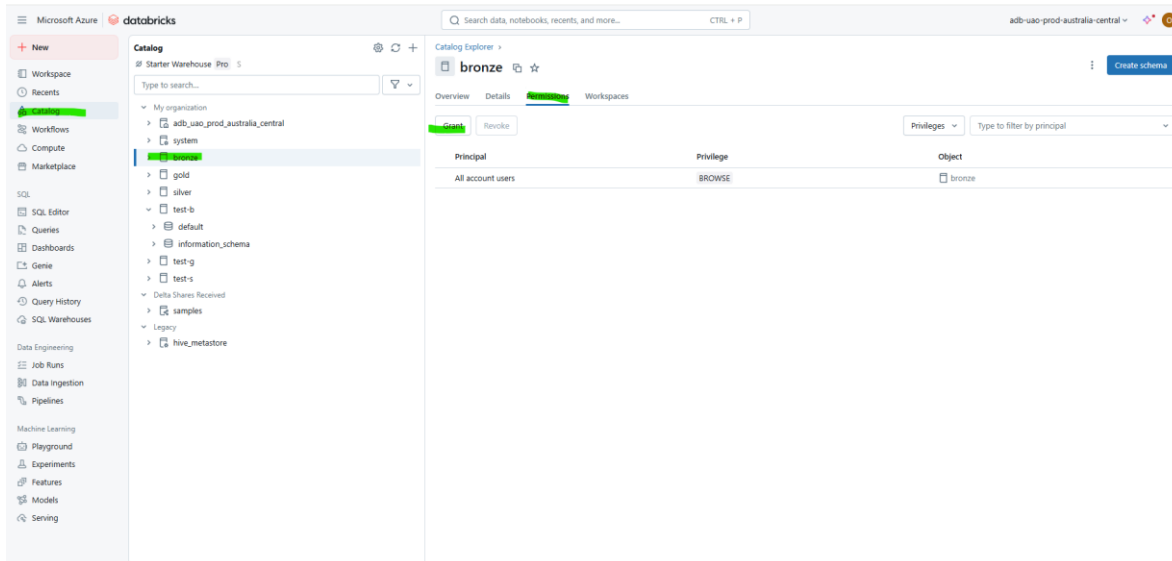
3. Damos clic en permissions, luego agregamos nuestro usuario, activamos los dos check box inferiores con los permisos y finalmente en grant.



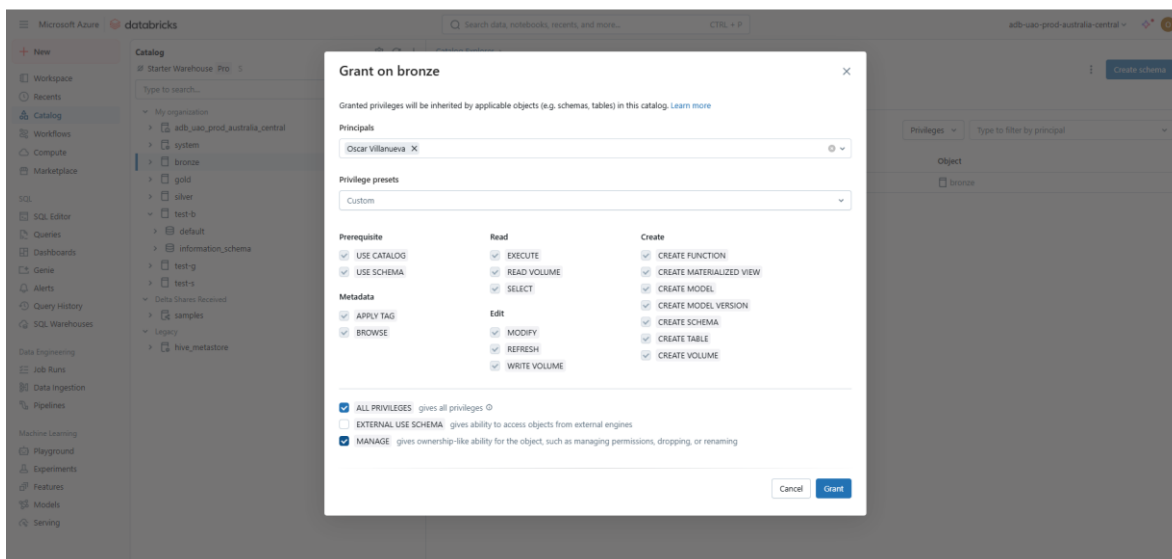
Otorgar permisos a los catálogos

Estos pasos se deben realizar por cada uno de los **3 catálogos** creados

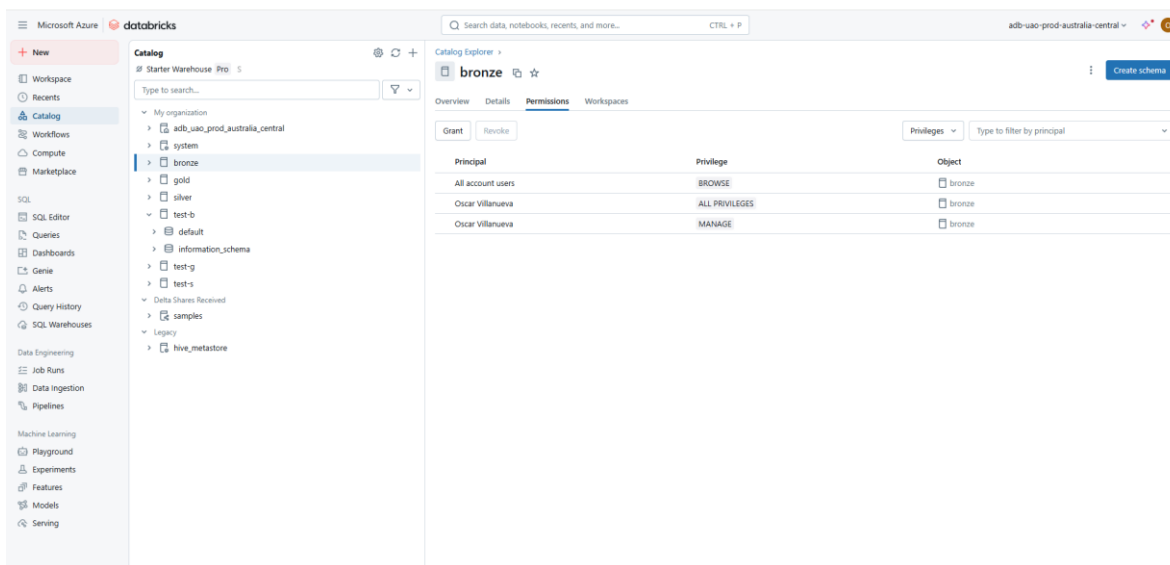
1. Desde Catalog , seleccionamos el catalogo a gestionar, en este caso empezamos con bronze. Luego seleccionamos Permissions y finalmente en grant



2. Ingresamos nuestro usuario y habilitamos los check box como se muestran en la imagen. Damos clic en grant.



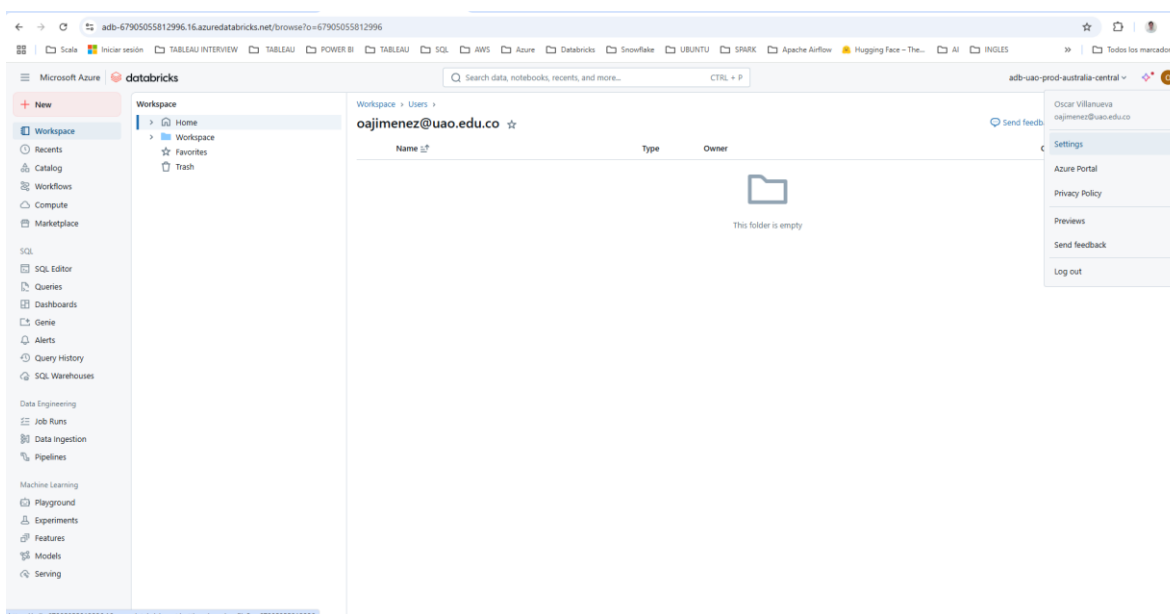
Queda así



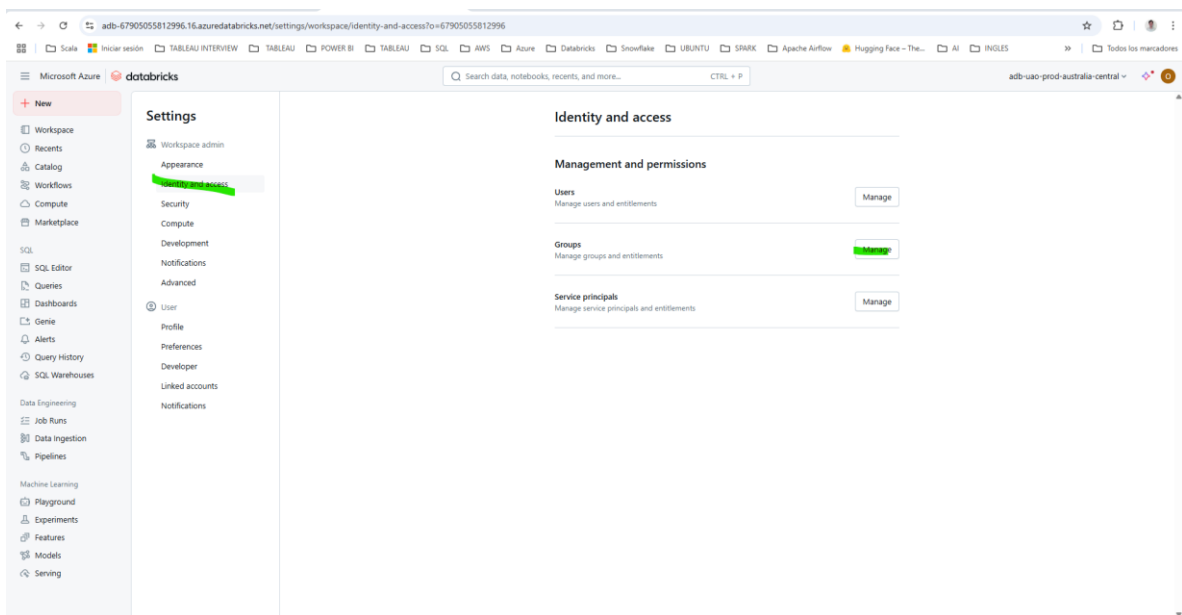
Definición de grupos de trabajo (opcional)

Todos los permisos tanto de credenciales, external location y catalog también pueden ser otorgados a nivel de grupo.

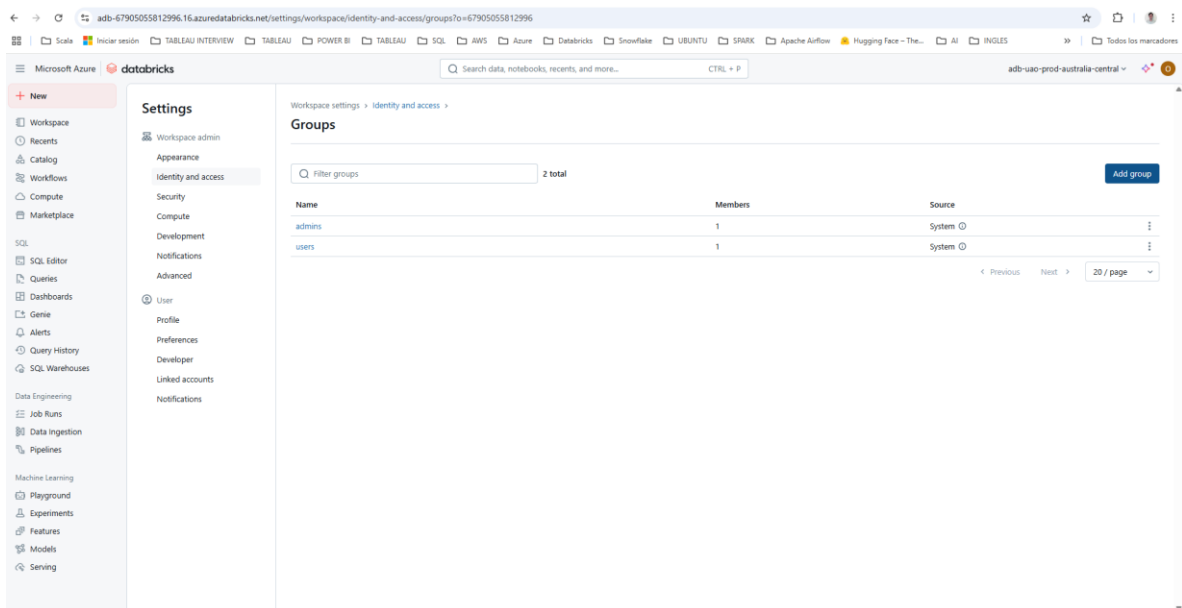
1. Desde settings



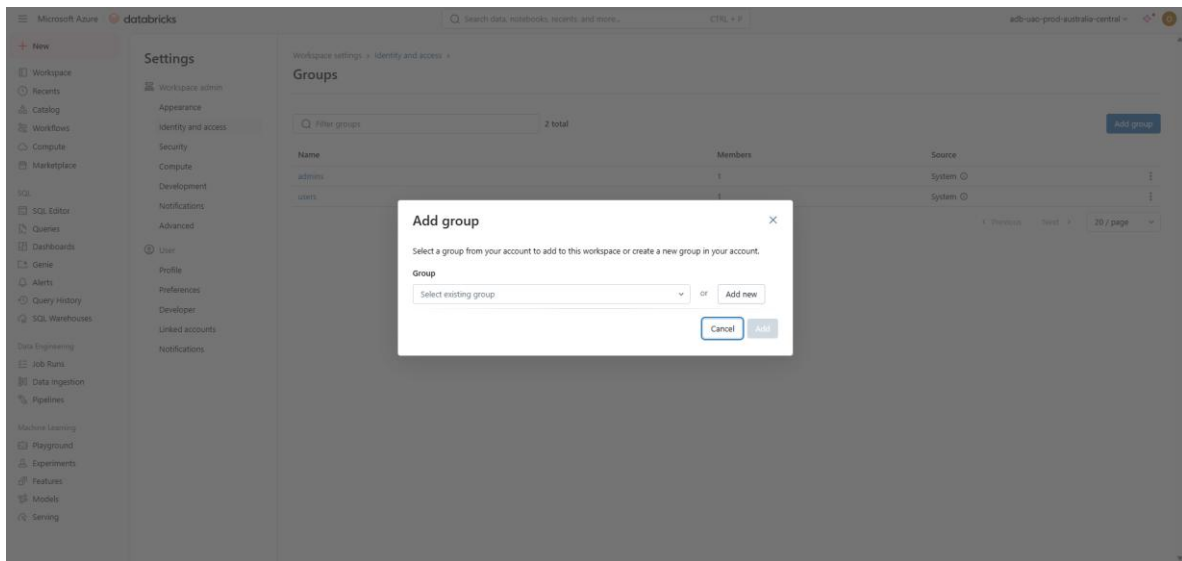
2. Ingresamos a identity and Access. Luego en groups y manage.



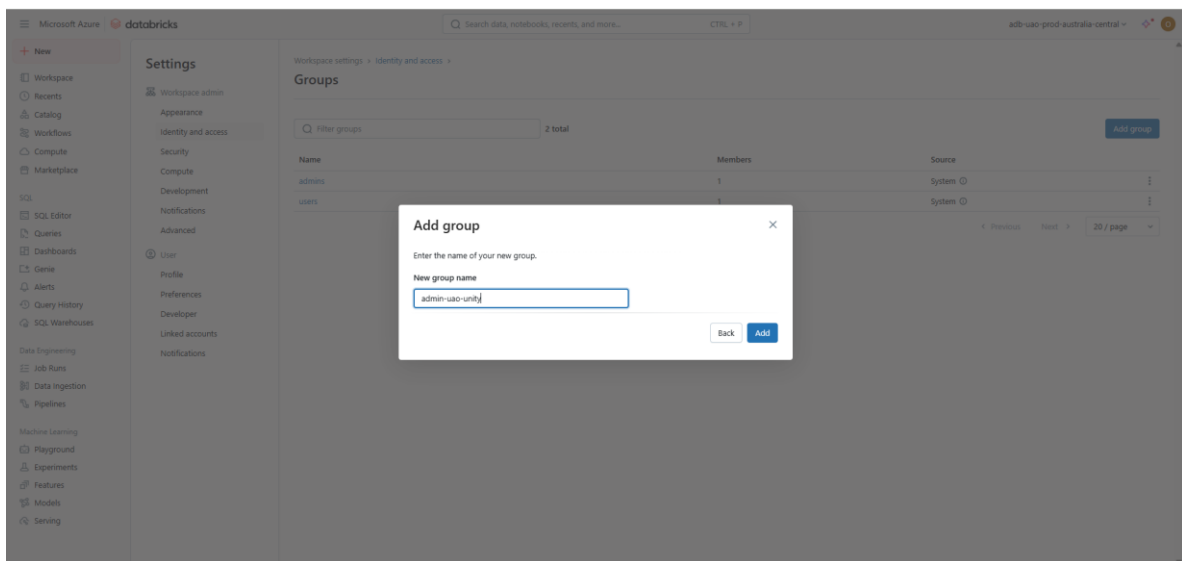
3. Adicionamos un grupo



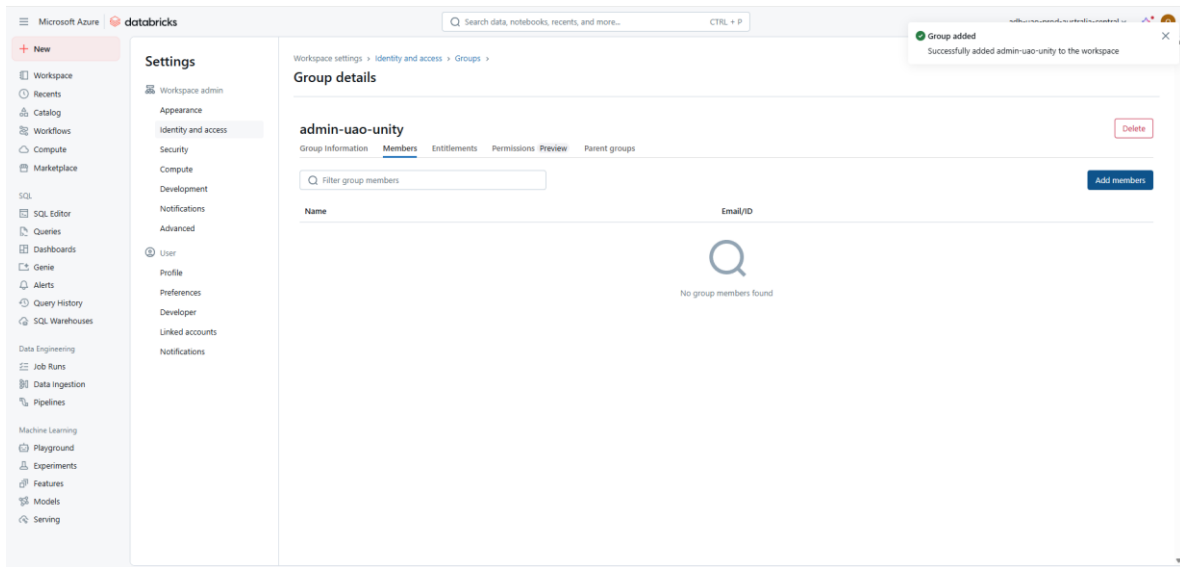
4. Agregamos uno nuevo



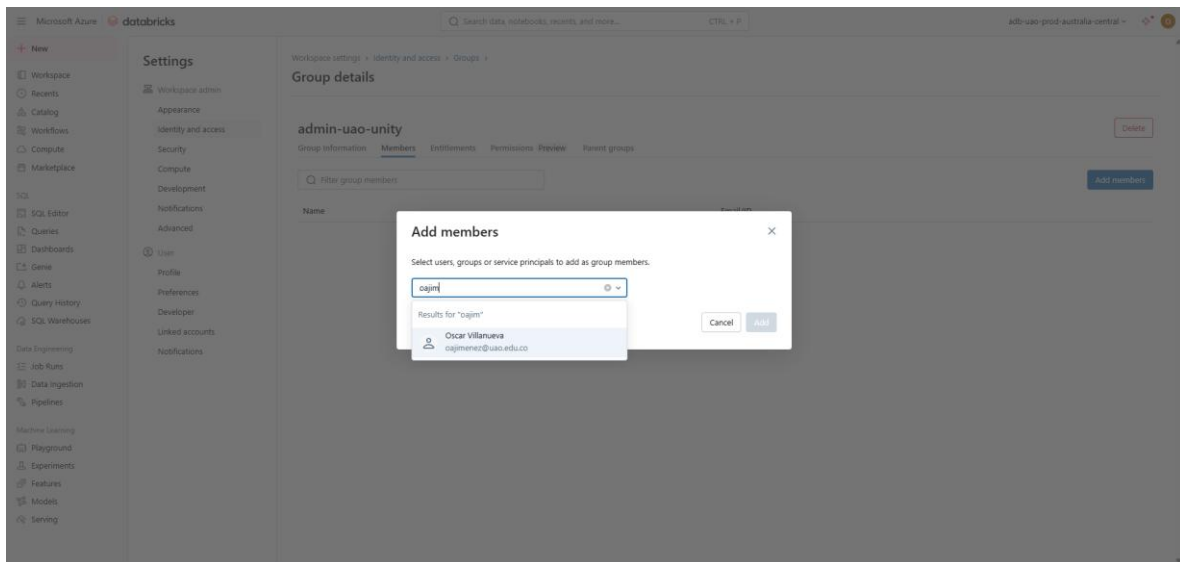
5.



6. Agregamos los integrantes que requiramos

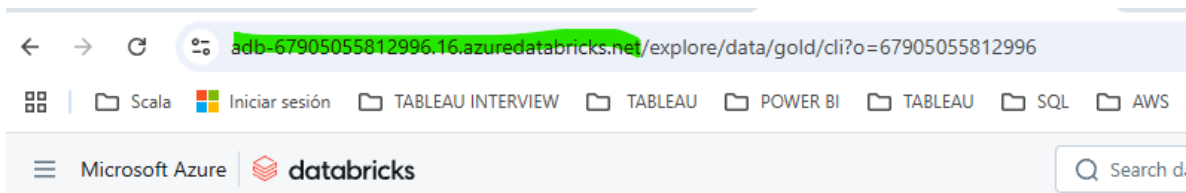


7. Agregamos los usuarios



Creación del Scope

1. Necesitamos definir esta url que extraemos desde nuestro databricks

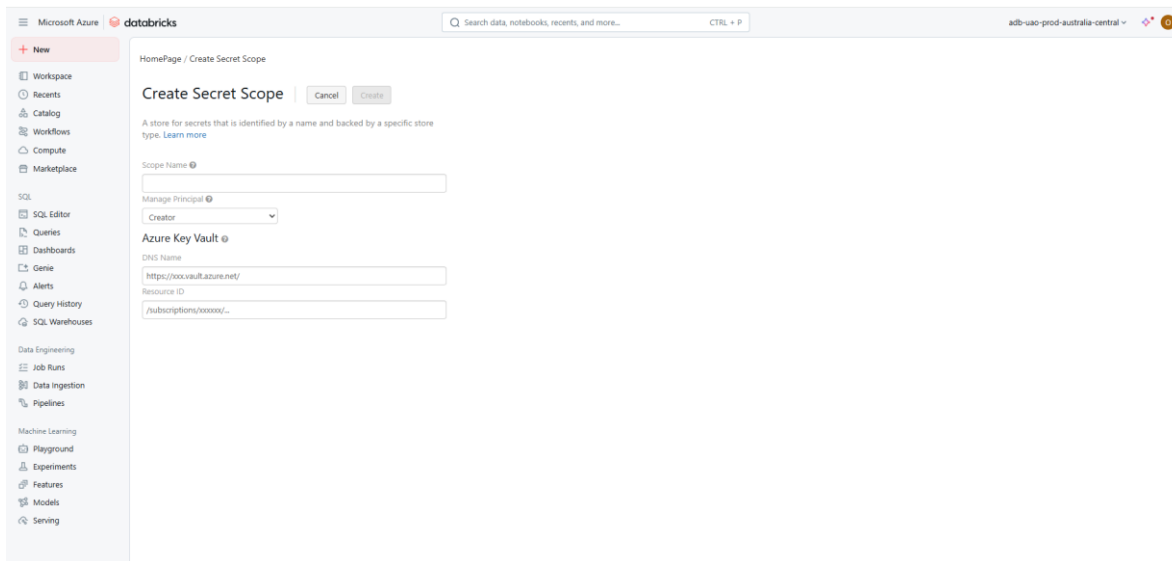


url/#secrets/createScope

Queda así

<https://adb-67905055812996.16.azure.databricks.net/#secrets/createScope>

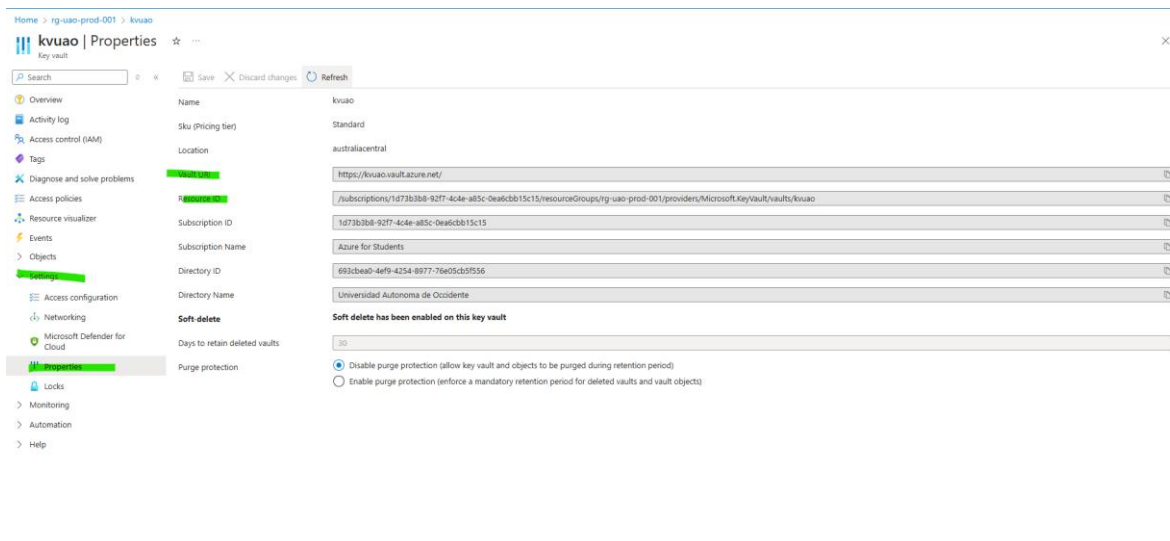
2. La pegamos en el navegador



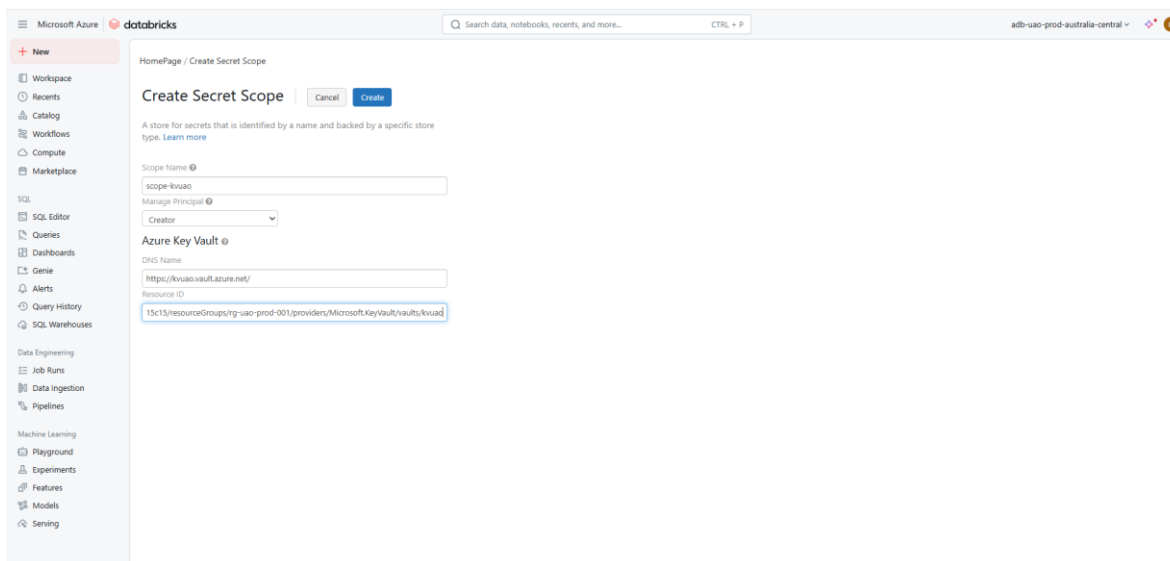
3. Definimos los parámetros como

- Scope name: scope-kvuao (es el nombre del key vault que desplegamos)
- DNS Name: Viene del campo vault URL en el key vault
- Resource ID: Viene del campo Resource ID en el key vault

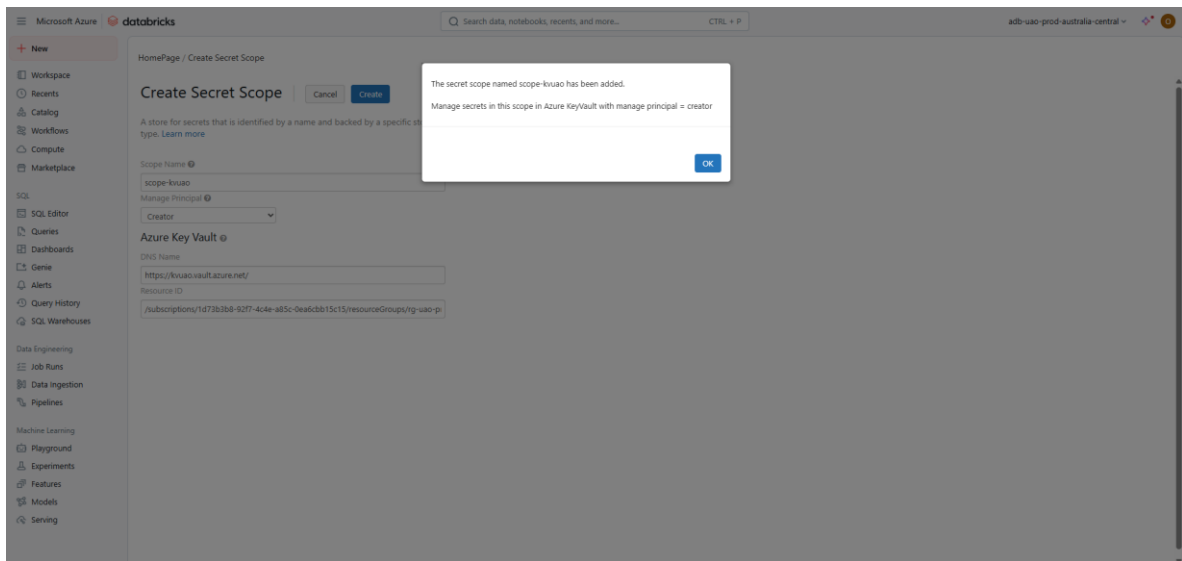
Ingresamos al key vault, en la pestaña settings, luego properties y copiamos los valores



4. Damos clic en crear

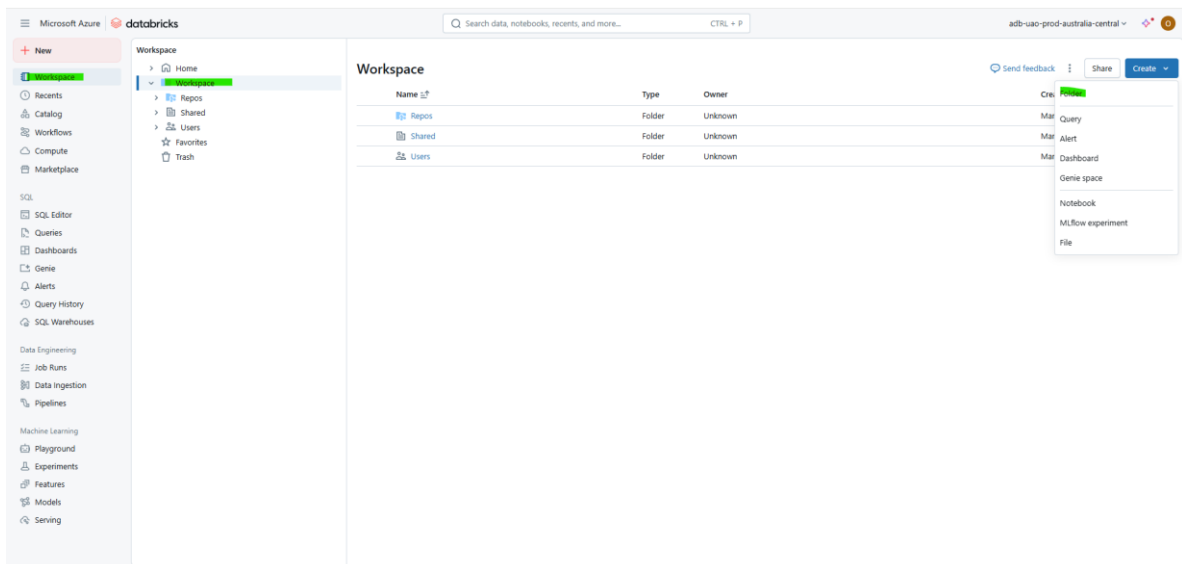


El scope permite que desde databricks se puedan leer todos los key vault.

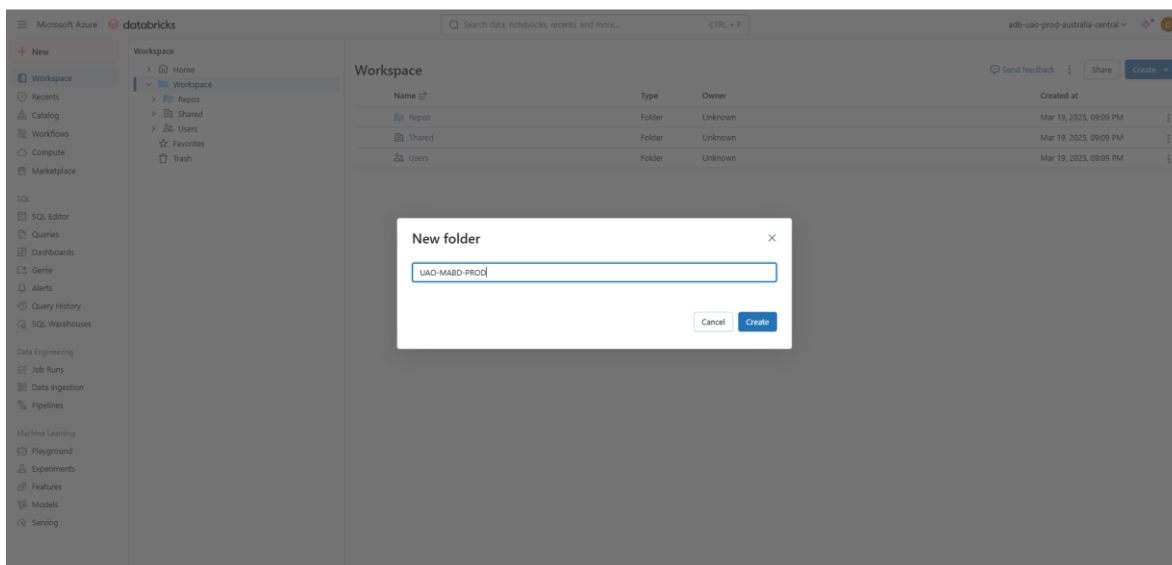


Creación de carpeta de trabajo

1. Desde workspace creamos una nueva carpeta de trabajo



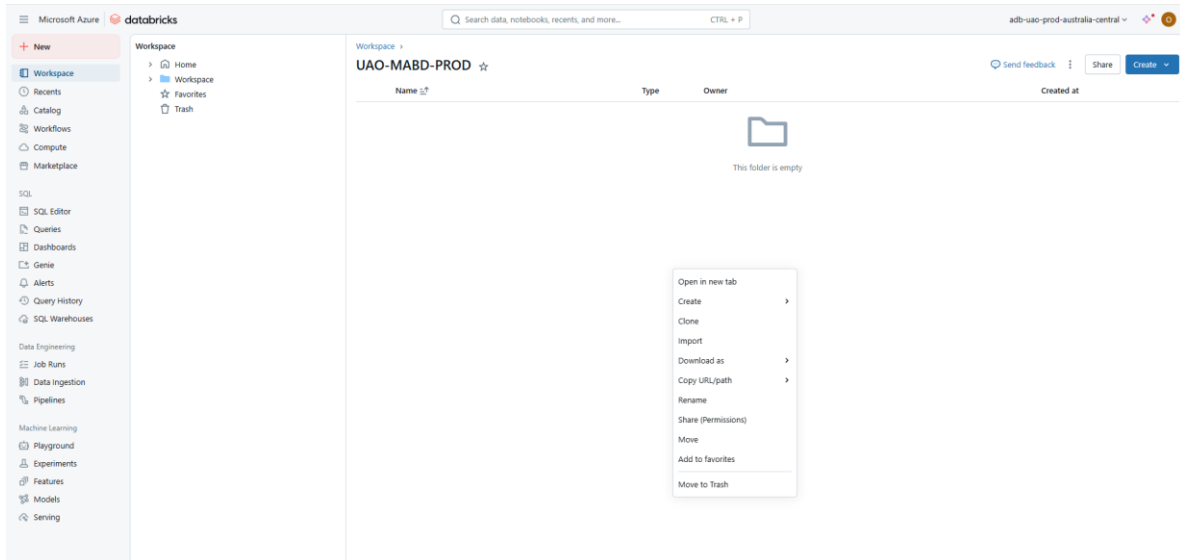
2. Definimos el nombre Y damos clic en create



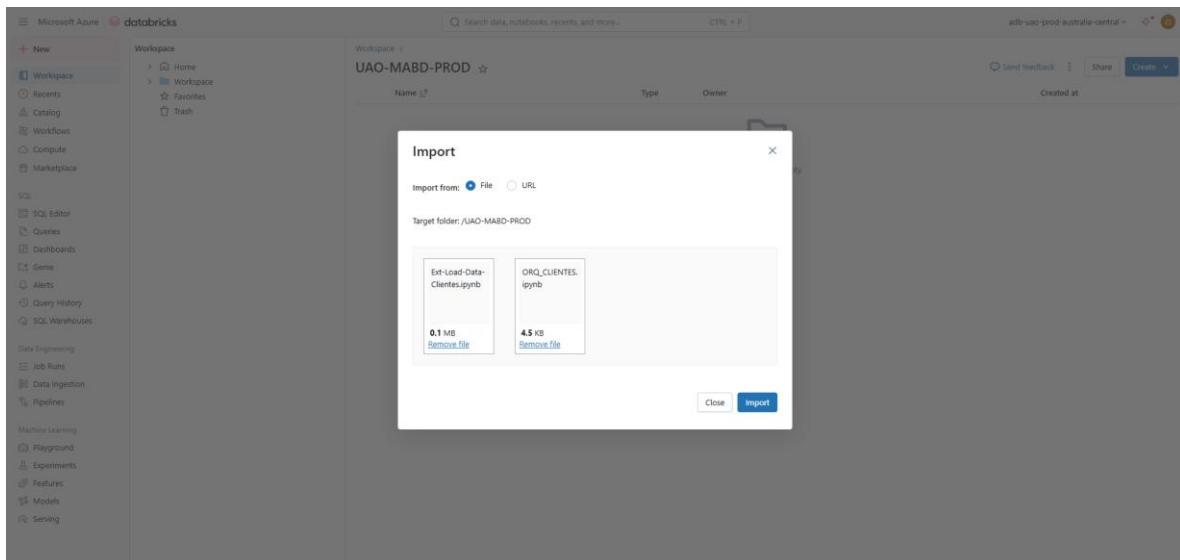
Parte 3

Importación de notebook de extracción y transformación

1. Desde nuestro folder damos clic derecho para abrir las opciones y damos clic en import



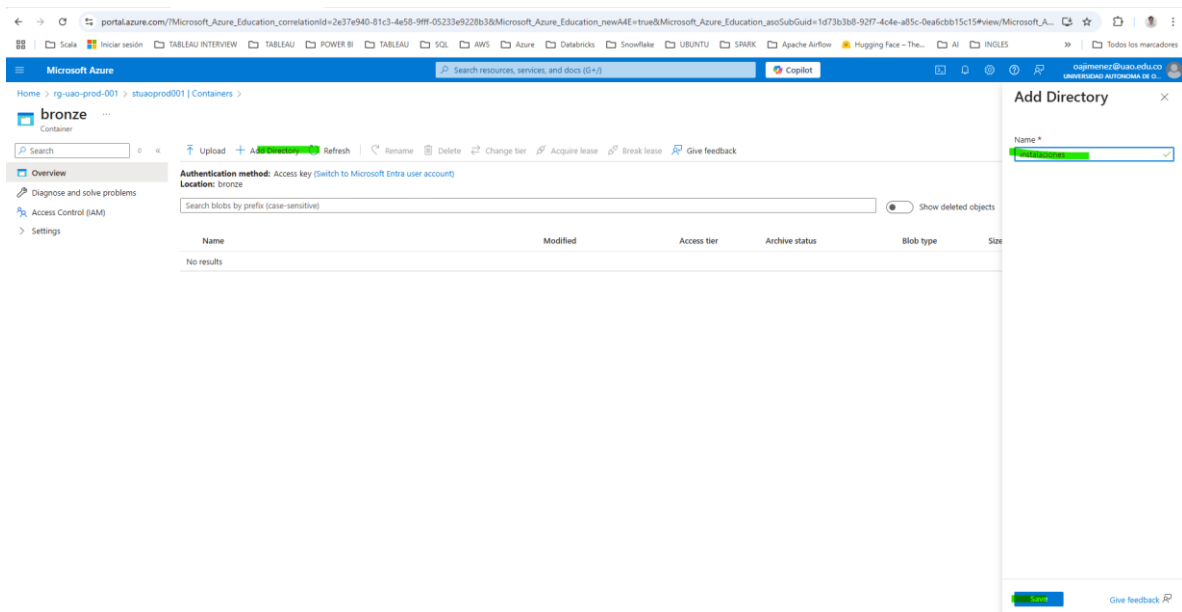
2. Cargamos los archivos de ejemplo



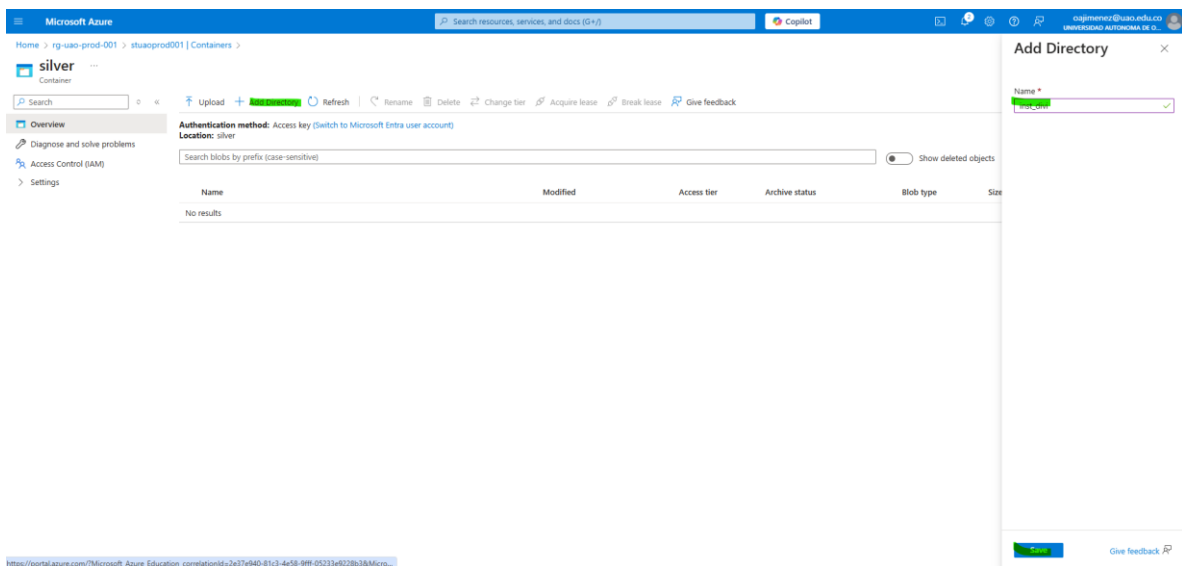
Creación de directorios para el almacenamiento de datos

1. Vamos a la cuenta de almacenamiento y dentro del contenedor de bronze creamos dos directorios:

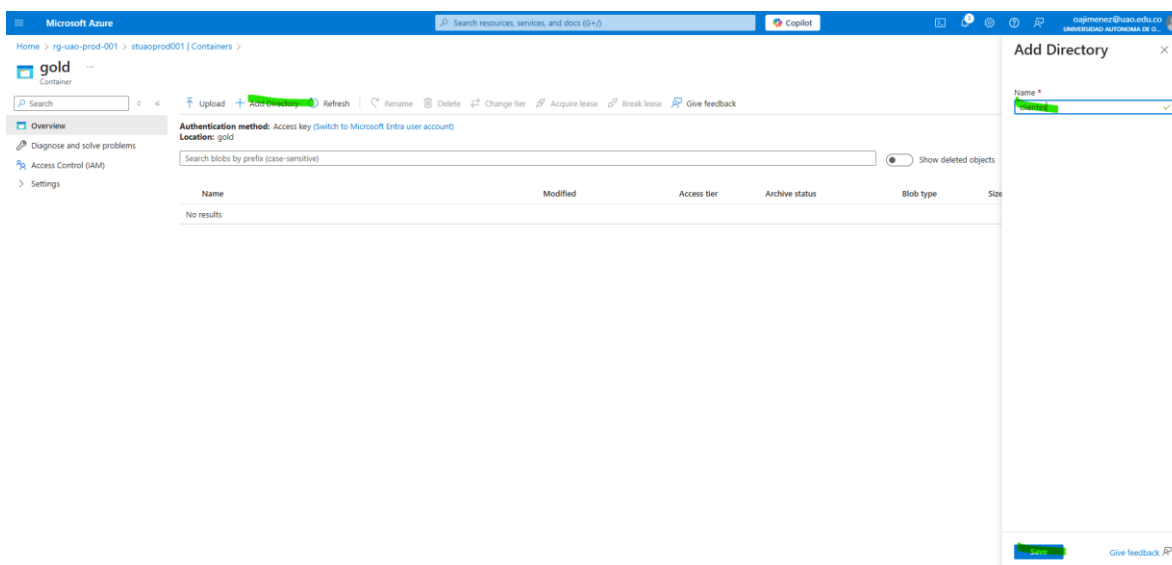
- instalaciones
- divipola



2. En el contenedor de silver creamos el directorio “inst_divi”

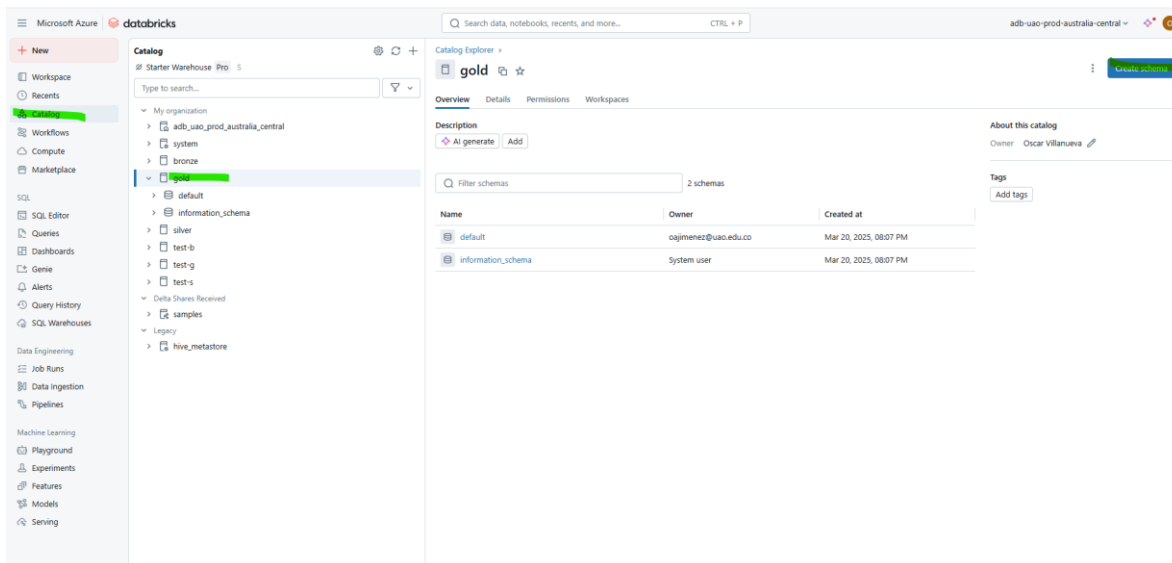


3. Finalmente, en el contenedor de gold creamos el directorio “clientes”

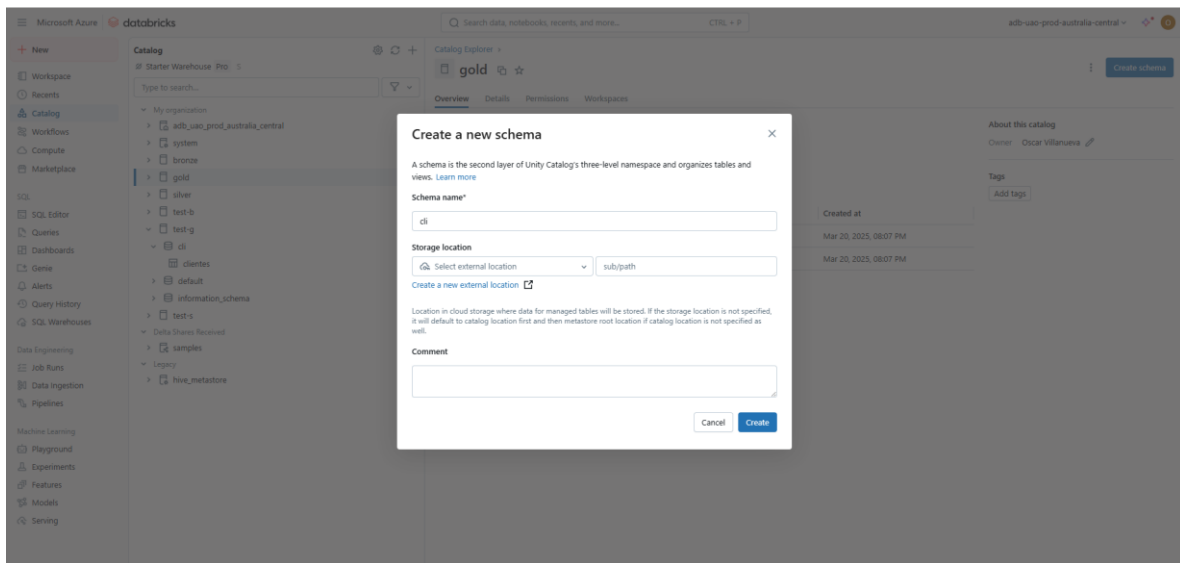


Creación de esquema de clientes

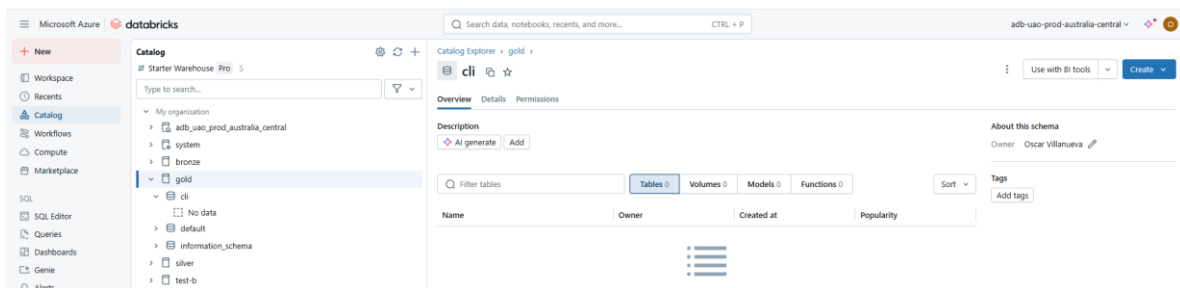
1. Desde Catalog seleccionamos nuestro catalogo gold y damos clic en crear esquema



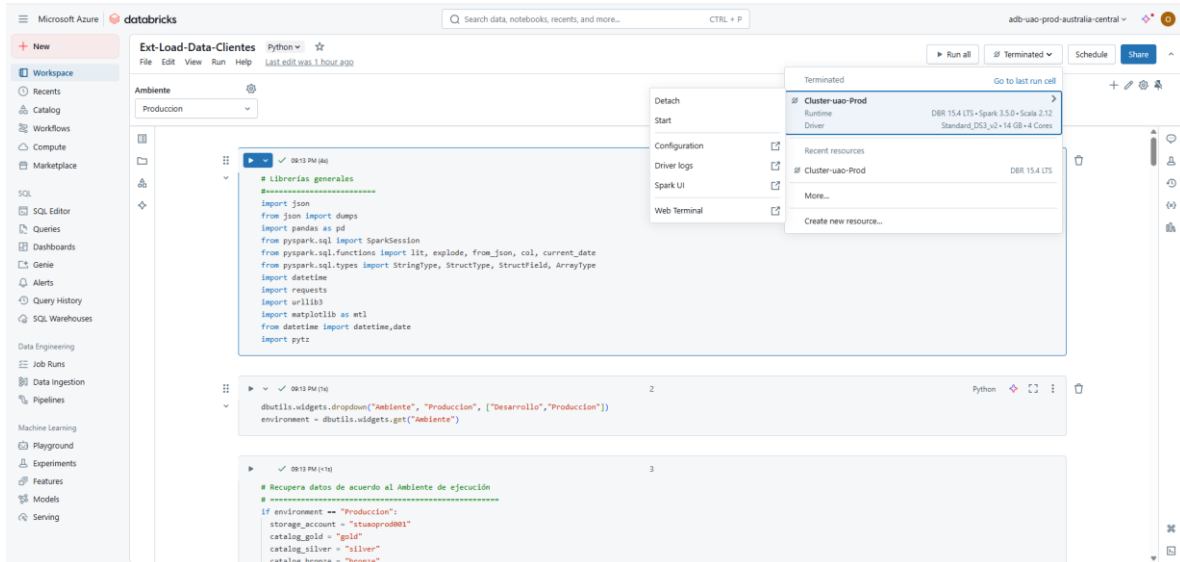
2. Definimos el nombre como “cli” y damos clic en create



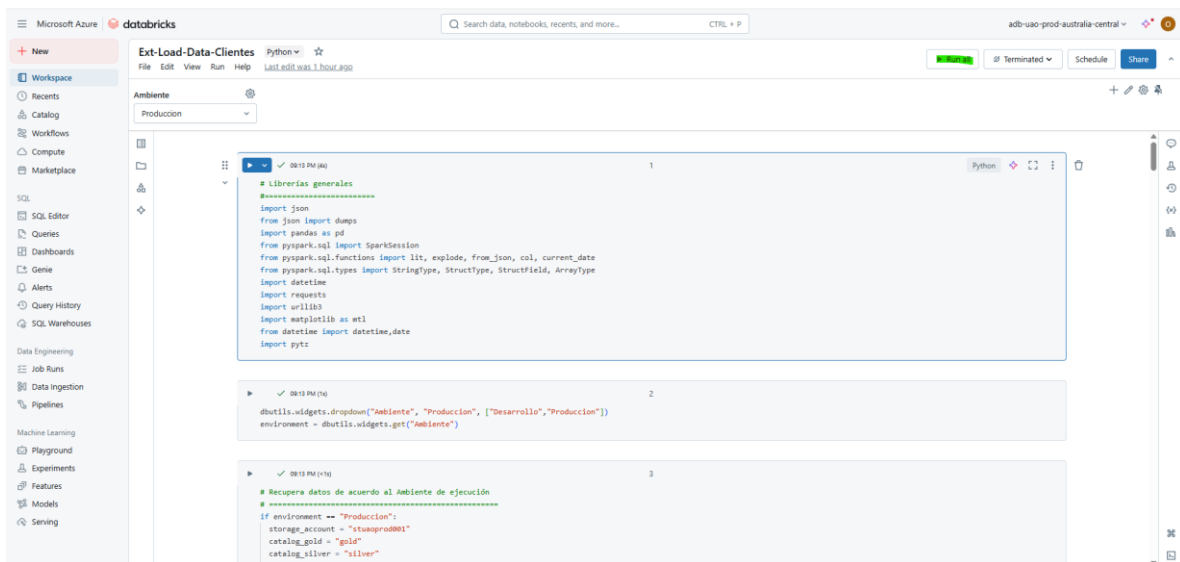
3. Ya queda definido nuestro esquema



4. Podemos ejecutar nuestro notebook. En caso de que esté apagado el cluster, lo seleccionamos y damos clic en start.

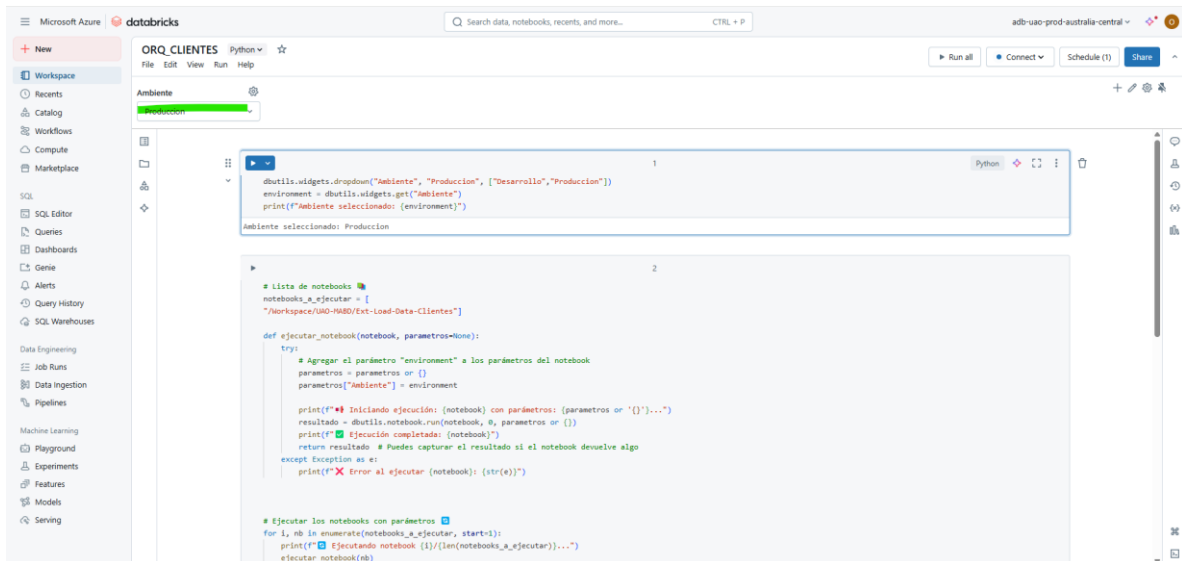


5. Finalmente, damos clic en Run all

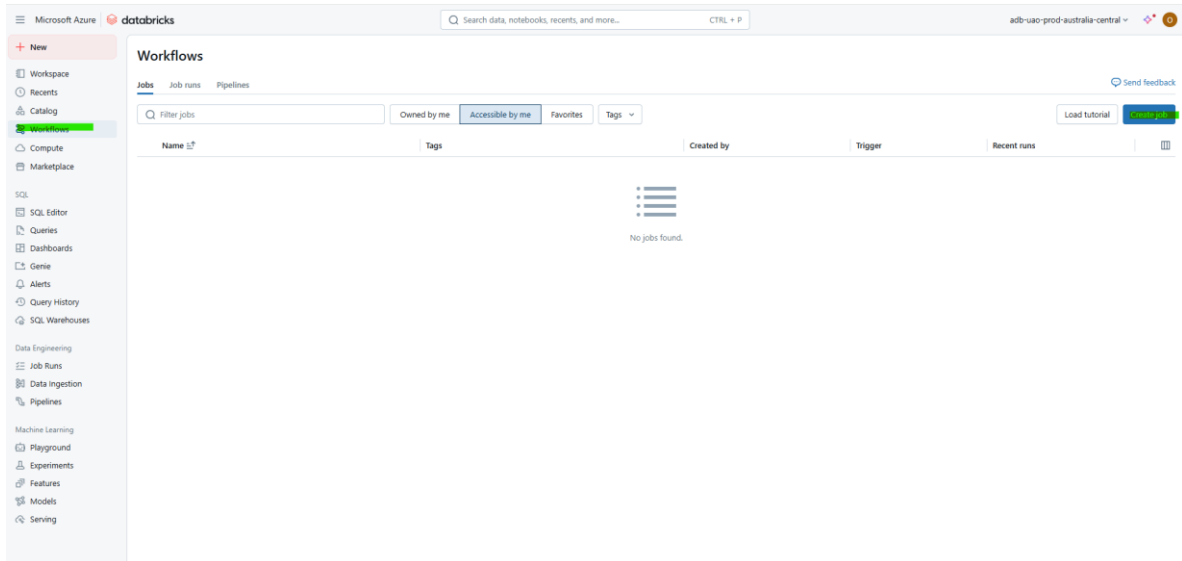


Programación de Job

Antes de iniciar el Job debemos ir al notebook ORQ_CLIENTES y seleccionar el ambiente producción:

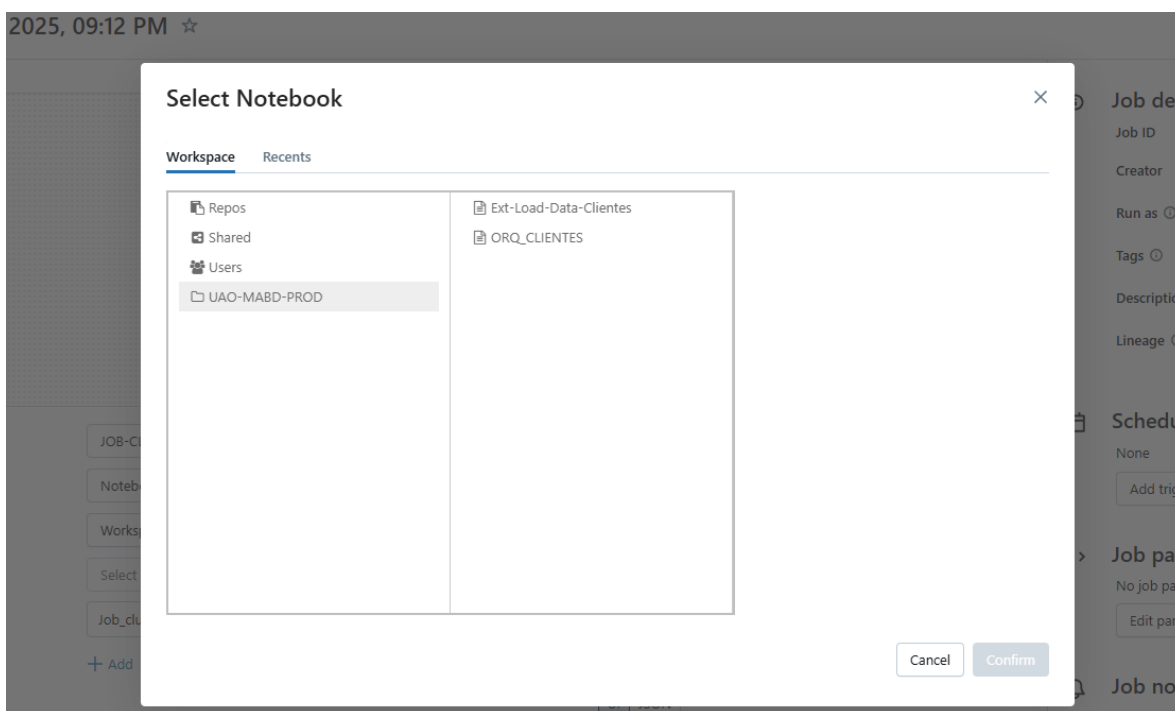


1. Desde workflows hacemos clic en create job



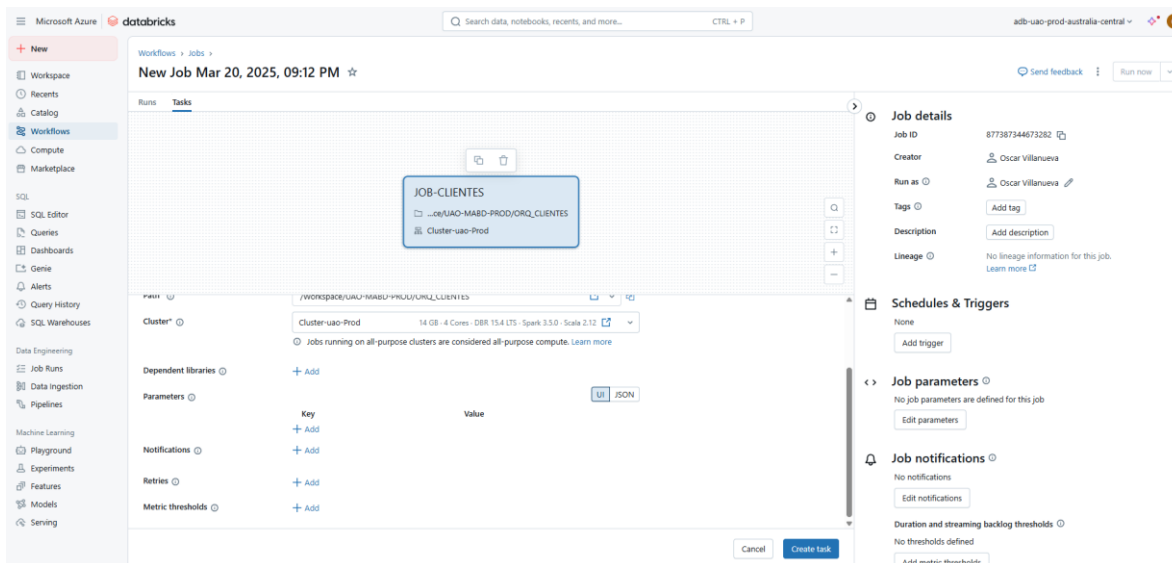
2. Definimos los siguientes parámetros:

- Name: JOB-CLIENTES
- Type: Notebook
- Source:Workspace
- Path: Seleccionamos la ruta donde se encuentra el archivo cargado previamente llamado ORQ-CLIENTES

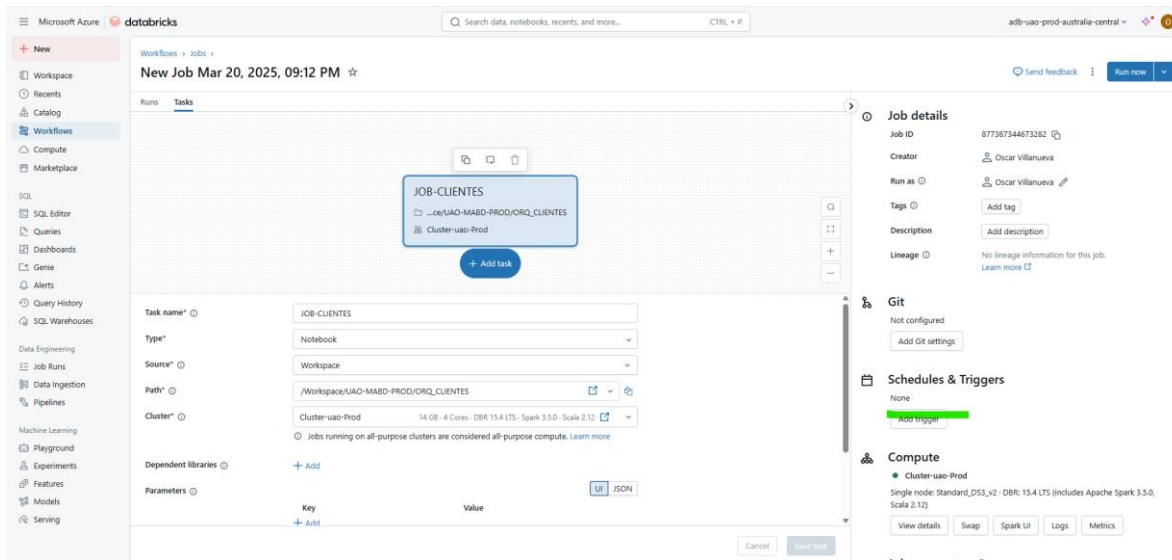


Cluster: elegimos el creado Clueter-uao-Prod.

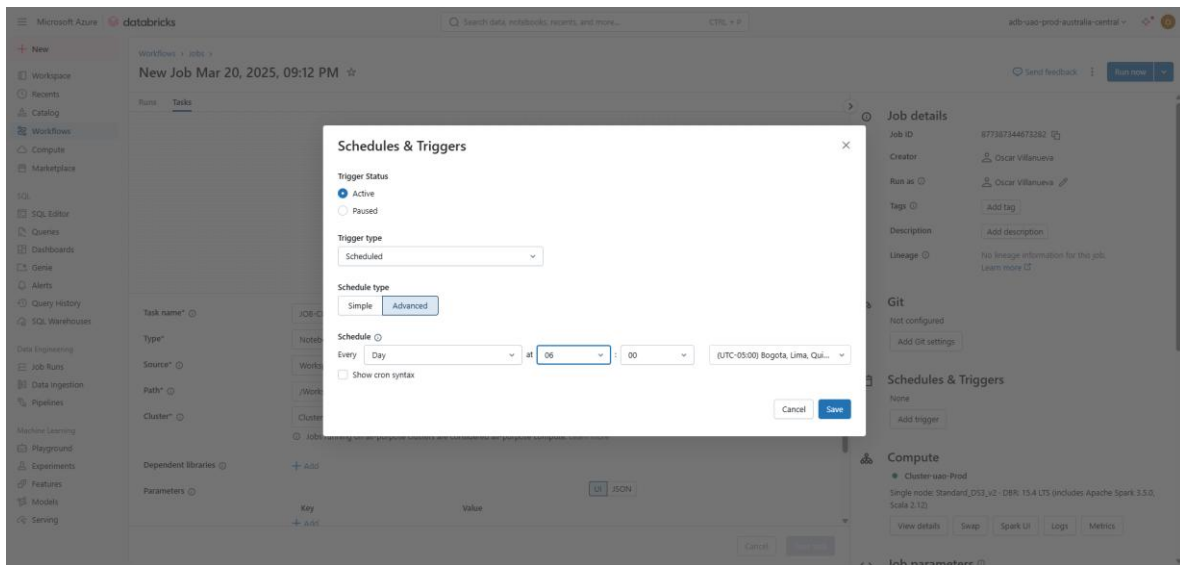
4. Damos clic en create task



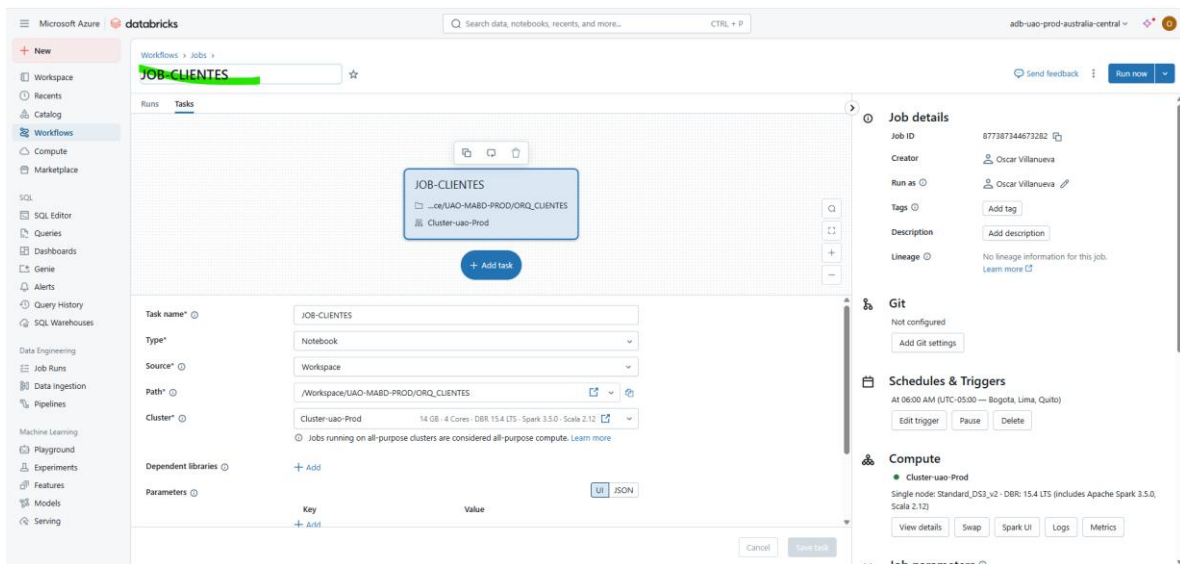
5. Definimos una ejecución programada desde add trigger



6. Definimos como scheduled y luego en advanced junto con la hora que queramos



7. Cambiar el nombre del job



8. Para probar el job vamos a Runs y corremos el job

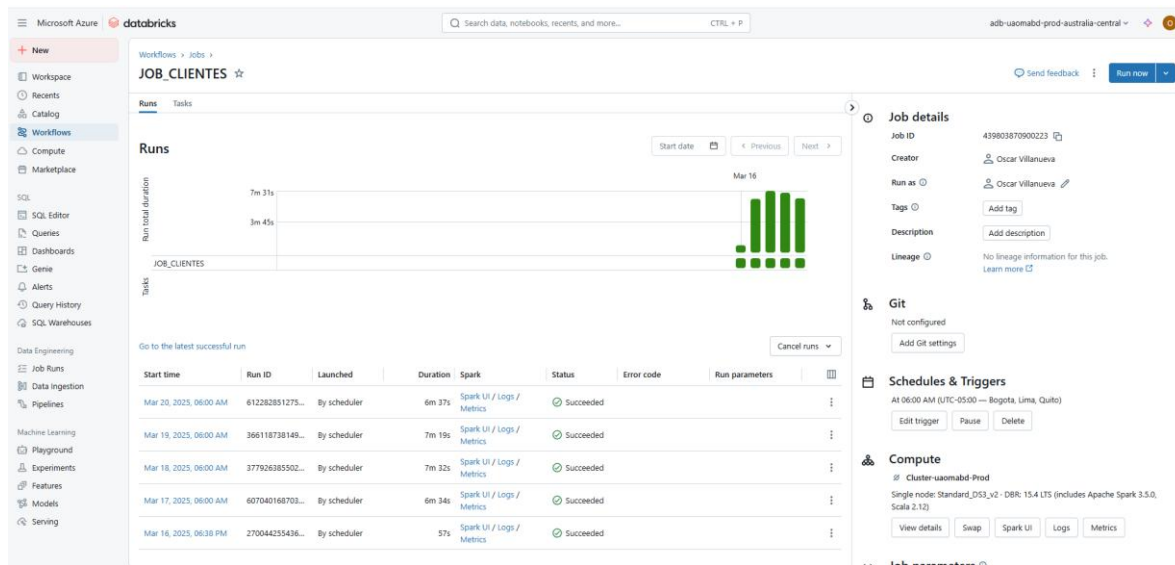
The screenshot shows the Databricks interface for a job named 'JOB-CLIENTES'. The left sidebar contains navigation options like Workspace, Recents, Catalog, Workflows, Compute, and Marketplace. The main area is divided into 'Runs' and 'Tasks' tabs. The 'Runs' tab displays a message: 'No job runs were found for this job within the last 60 days'. The 'Job details' panel on the right provides information about the job, including its ID (877387344673282), creator (Oscar Villanueva), and a 'Run now' button. Other details include 'Run as', 'Tags', 'Description', 'Lineage', 'Git' status, and 'Schedules & Triggers'.

9. El job se ejecutó correctamente

The screenshot shows the Databricks interface for the same job 'JOB-CLIENTES'. The 'Runs' tab now displays a bar chart showing the run duration and a table of runs. The table has columns for Start time, Run ID, Launched, Duration, Spark, Status, Error code, and Run parameters. The first row shows a successful run on Mar 20, 2025, at 09:24 PM, with Run ID 718247314108..., launched manually, with a duration of 27s, and a status of 'Succeeded'. The 'Job details' panel on the right remains the same, showing job ID 877387344673282 and the 'Run now' button.

Start time	Run ID	Launched	Duration	Spark	Status	Error code	Run parameters
Mar 20, 2025, 09:24 PM	718247314108...	Manually	27s	Spark UI / Logs / Metrics	Succeeded		

10. Así se ve una ejecución diaria exitosa del Job



FIN



Algunos recursos adicionales:

- <https://azuremarketplace.microsoft.com/en-us/marketplace/apps/microsoft.accessconnector?tab=overview>
- <https://learn.microsoft.com/en-us/azure/databricks/data-governance/unity-catalog/azure-managed-identities>
- <https://learn.microsoft.com/es-es/azure/key-vault/general/basic-concepts>
- <https://learn.microsoft.com/en-us/azure/databricks/sql/language-manual/sql-ref-external-locations>
- <https://learn.microsoft.com/en-us/azure/databricks/security/secrets/>